Fresenius Kabi USA, LLC
Three Corporate Drive
Lake Zurich, IL 60047

1–800–937–5060

Technical Information Document

DXT Data Management System Data Interface for AURORA and AURORA Xi

The purpose of this document is to define the interface specification required for communication between DXT and external Data Management System (DMS).

This specification defines the physical interface, data communication protocol, and data content of messages passed between DXT and the DMS.

Table of Contents

Chapter 1 General Description 4
Section 1.1 Overview 4
Section 1.2 Web Services 4
Procedure Parameters 4
Procedure Parameter Status 7
Electronic Record 7
Instrument Status 7
Section 1.3 Message Format 7
HTTP Request 7
Section 1.4 Root Element 8
Chapter 2 DXT Server 9
Section 2.1 Procedure Parameters 9
Procedure Parameters URL 9
Procedure Parameters Content Type 9
ProcedureParameters XSD Definition 9
ProcedureParameters Data Description 10
ProcedureParameters DXT Response 11
Section 2.2 Procedure Parameters for AURORA Instruments 12
DXTProcedureParameters XSD Definition 12

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

DXTProcedureParameters Data Description 14
Procedure Parameters Example for AURORA Instruments 15
Section 2.3 Procedure Parameters for AURORA Xi Instruments 16
DXTProcedureParameters XSD Definition 16
DXTProcedureParameters Data Description 19
Procedure Parameters Example for AURORA Xi Instruments 23
Section 2.4 Instrument Status 24
Instrument Status URL 24
Instrument Status DXT Response 24
InstrumentStatus XSD Definition 24
InstrumentStatus Data Description 26
Instrument Status Response Example 28
Section 2.5 DXT Server Response 28
OK Response 28
Request Header Validation Error Response 29
Request Time Validation Error Response 30
Request Length Validation Error Response 30
Request HTTP Version Validation Error Response 31
Request Processing Error Response 31
Chapter 3 DXT Client 33
Section 3.1 Electronic Records for AURORA and AURORA Xi instruments 33
Electronic Record URL 33
ElectronicRecord XSD Definition 33
ElectronicRecord Data Description 36
DXTProcedureEvents XSD Definition 41
DXTProcedureEvents Data Description 43
Custom Events 47
DXTProcedureTimespans XSD Definition 47
DXTProcedureTimespans Data Description 48
DXTProcedureDataDonation XSD Definition 49

Fresenius Kabi USA, LLC
All rights reserved

DXTProcedureDataNonDonation XSD Definition 52
DXTProcedureDataDonation and DXTProcedureDataNonDonation Data Description 54
DXTProcedureSoftGoods XSD Definition 58
DXTProcedureSoftGoods Data Description 58
DXTProcedureProducts XSD Definition 60
DXTProcedureProducts Data Description 61
Electronic Record Example for Donation Record 63
Electronic Record Example for Non–Donation Record 77
Section 3.2 Procedure Parameter Status 80
Procedure Parameter Status URL 80
ProcedureParameterStatus XSD Definition 80
ProcedureParameterStatus Data Description 81
ProcedureParameterStatus Example 82
Section 3.3 DMS Response 82
OK Response 82
Error Response 82

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Chapter 1 General Description

DXT is built using REST principles which ensure predictable URLs that make writing interface easy. The DXT interfaces follow HTTP rules, enabling a wide range of HTTP clients and servers to be used to interact with DXT.

This chapter gives an overview of DXT interface structure.

- Overview
- Web Services
  - Procedure Parameters
  - Procedure Parameters Status
  - Electronic Record
- Message Format
  - HTTP Request
  - HTTP Response
  - Root Element

Section 1.1 Overview

DXT implements bi–directional communications with external data management systems (DMS) to provide asynchronous communications. Based on the type of the message, the communications can be initiated by either DXT or the DMS.

For example, when the DMS sends procedure setup file to DXT, the DMS will act as a client posting setup data to the DXT server. Likewise, when DXT sends an electronic record to the DMS, DXT will act as a client posting procedure data to the DMS server.

Section 1.2 Web Services

This section provides an overview of the web services included with DXT. The following chapters contain detailed description of each service as well as structure and content of each message.

Procedure Parameters

The DMS will use the Procedure Parameters service to send procedure setup file to DXT.

The procedure setup parameters include details about the donor to enable the instrument to determine the appropriate procedure settings for the donor.

All rights reserved

Status CURRENT Effective 8/13/2019

The main purpose of remote procedure setup is to help improve procedure setup time and help reduce manual entry errors at the instrument.

There are two different methods available to achieve this, the push method and pull method.

Push Method

In the push model of remote procedure setup, the DMS will send the instrument status request to determine availability of the instruments for the remote procedure setup. The DMS will select an instrument and send the procedure setup file to DXT, then DXT will send the procedure setup file to the instrument specified by the DMS. This is commonly referred to as the push model since DXT is pushing the data down to the instrument.

The following sequence diagram depicts interactions between the DMS, DXT, instrument, and operator for the push model.

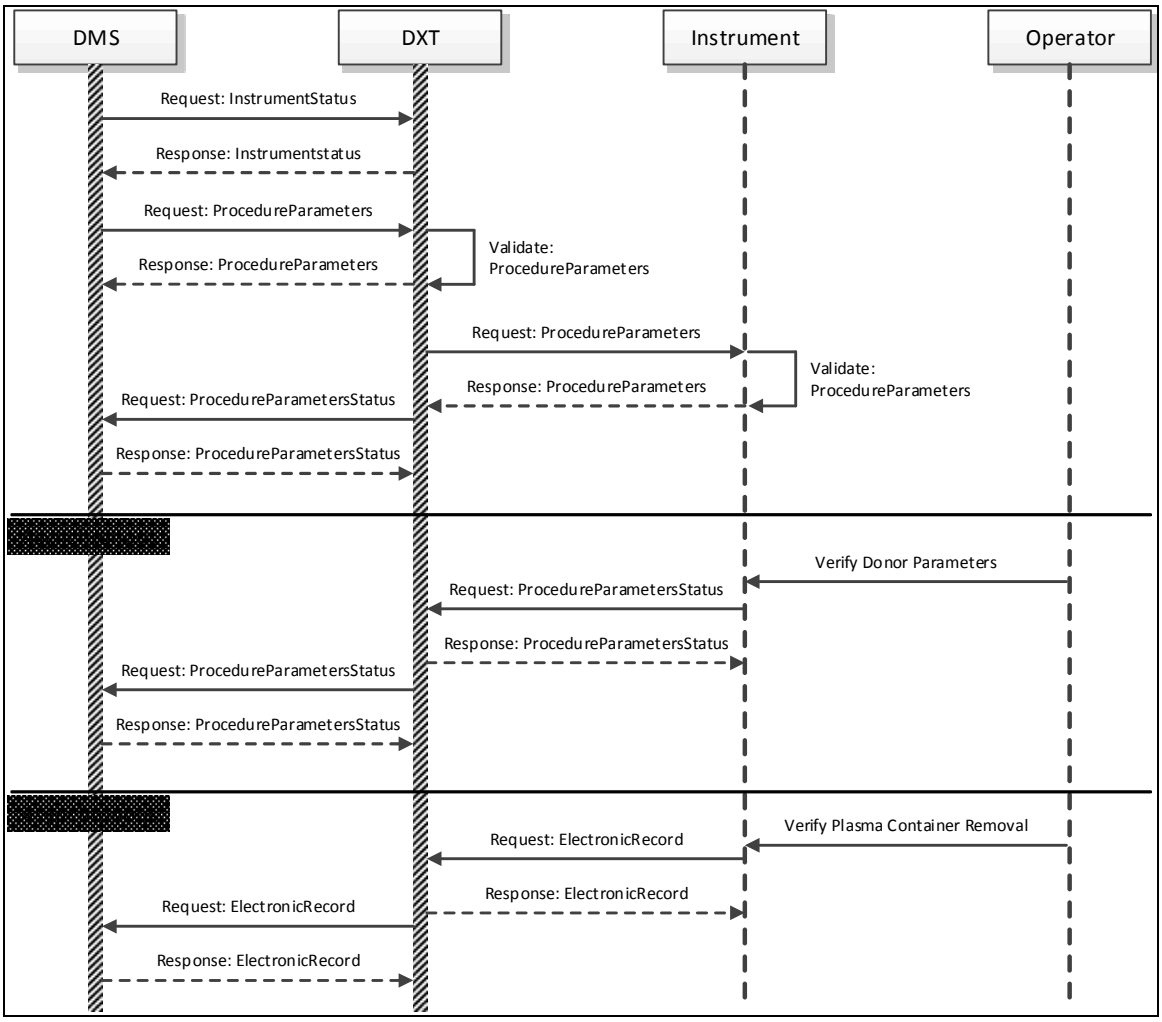


Figure 1 Typical Sequence of Interactions in PUSH Mode

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Pull Method

In the pull model of remote procedure setup, the DMS provides the procedure setup file to DXT. DXT will queue the information, making it visible in DXT in the Monitoring feature, on the Donation Setup tab. When an operator enters or scans the Donation ID or Donation Setup ID into the apheresis instrument, the instrument triggers a request for the procedure seup file from DXT. This is commonly referred to as the pull model since the instrument is pulling the data from DXT.

The ollowing sequence diagram depicts interactions between DMS, DXT, instrument, and operator for the pull model.

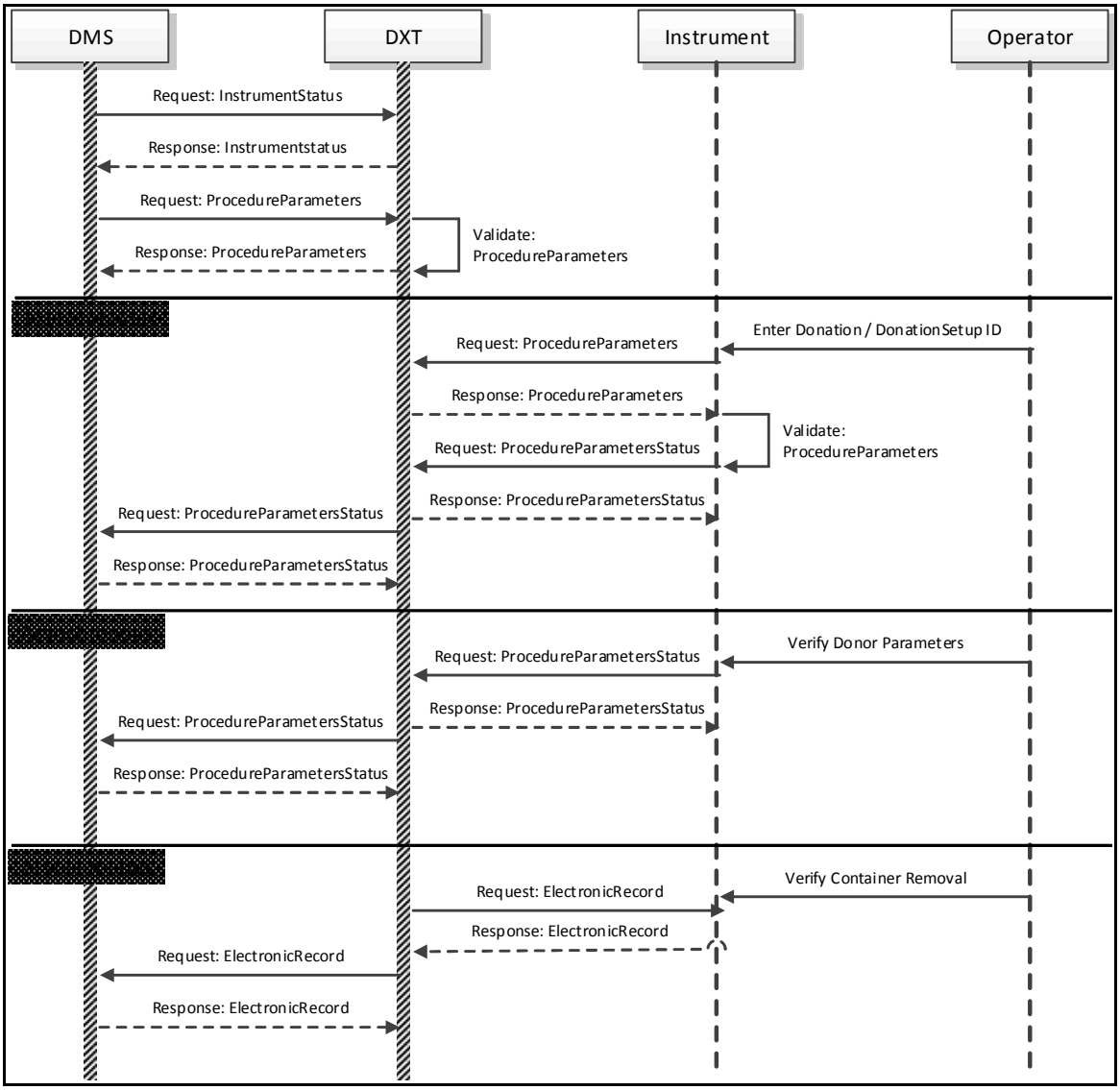


Figure 2 Typical Sequence of Interactions in PULL Mode

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Procedure Parameter Status

DXT will use the Procedure Parameter Status service to notify the DMS of the status of the procedure parameter.

Electronic Record

DXT will use the Electronic Record service to provide the procedure record data to the DMS.

DXT receives and stores data from connected instruments. Using the Electronic Record service, DXT can forward the data to the DMS. The data maintained in DXT can be considered as the 21 CFR Part 11 and/or EU Annex 11 compliant electronic records for a blood or platelet center, provided the center has policies and procedures for system usage as described in the regulation. Refer to DXT Data Management System Administrator's Guide Chapter 7 – Cyber Security, Part 11, and Annex 11 for considerations for deploying DXT in a 21 CFR Part 11 and/or EU Annex 11 compliant environment.

Instrument Status

The DMS will use the Instrument Status service to determine the availability of the site's instruments for remote procedure setup.

Section 1.3 Message Format

HTTP Request

When DXT acts as a client, it will send all HTTP request headers in the following format. When DXT acts as a server, it expects all HTTP request headers in the same format.

```
METHOD URL HTTP/1.1
Host: SERVER_ADDRESS
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Connection: Keep-Alive
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

Method must be either GET or POST, DXT does not currently support resources updates (PUT) or deletion (DELETE). DXT will return an error response to HTTP PUT and HTTP DELETE requests.

HTTP header **Host** must be present and contain network name of the destination server.

HTTP Header **Content–Length** must be present and contain payload (XML) length in bytes.

HTTP Header **Content–Type** must be "text/xml; charset=utf–8", unless otherwise noted for the specific message.

HTTP Header **Date** must be present and contain date and time when message was generated using full Date format, specified by RFC–2616, Hypertext Transfer Protocol –– HTTP/1.1. For example:

All rights reserved

```
Date: Fri, 14 Jun 2013 21:12:59 GMT
```

DXT expects all messages to be received within 2 minutes to ensure proper functionality. DXT will respond with an error to any message that contains an HTTP header Date Time more than 2 minutes in the past or in the future from DXT server time.

HTTP Header must include **Connection: Keep–Alive** directive.

HTTP Header must include **Cache–Control** with cache directive set to “no–cache” and “no–transform”. Comma “,” and additional values (e.g., “private”) are acceptable.

The value of HTTP Header **FK–SourceId** and **FK–DestinationId** must be set as following:

For DXT, it should be “0”.
For DMS, it should be “1”.
For QC System, it should be “2”.

DXT expects **FK–SourceId** in HTTP Header of all inbound messages, and will supply **FK–DestinationId** in HTTP Header of all outbound messages.

For example, when the DMS sends data to DXT as a server, the IDs will be as following:

```
FK-SourceId: 1
FK-DestinationId: 0
```

Section 1.4 Root Element

All XML messages must have a root element of DMSMessage as shown below. No data is associated with this element, it is a container for all child elements.

```
<xsd:element name="DMSMessage">
    <xsd:complexType>
        <xsd:sequence>

            <CHILD ELEMENTS>

            </xsd:sequence>
</xsd:complexType>
</xsd:element>
```

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Chapter 2 DXT Server

This chapter defines messages that DXT accepts from external data management systems.

For possible error responses from DXT refer to Error Handling section at the end of this chapter.

Section 2.1 Procedure Parameters

The DMS will use the Procedure Parameters service to send procedure parameters to DXT. This enables the remote procedure setup if DXT is licensed for this feature.

Procedure Parameters URL

```
POST /fresenius-kabi/dxt/dms/procedure-parameters HTTP/1.1
```

Procedure Parameters request to DXT must supply ProcedureParameters XML object.

Procedure Parameters Content Type

Procedure Parameters request to DXT must contain the following content type header element:

```
Content-Type: multipart/mixed;
boundary=this_is_fresenius_kabi_http_boundary
```

The first part of the message must have content type “text/xml; charset=utf–8” and XML parameters file. Other parts are reserved for future use and should not be supplied:

```
--this_is_fresenius_kabi_http_boundary
Content-Type: text/xml; charset=utf-8

<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <ProcedureParameters>

        ...

    </ProcedureParameters>

</DMSMessage>
--this_is_fresenius_kabi_http_boundary--
```

ProcedureParameters XSD Definition

This complex type defines the contents of the Procedure Parameters structure as shown below.

```
<xsd:complexType name="ProcedureParameters">
    <xsd:sequence>

        <xsd:element name="DonationSetupId" minOccurs="1" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
                    <xsd:pattern value="[a-zA-Z0-9]*" />
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="InstrumentSerialNumber" minOccurs="0">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="20" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="SiteIdentifier" minOccurs="1" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="20" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DXTProcedureParameters" type="xsd:anyType" />

    </xsd:sequence>
</xsd:complexType>
```

ProcedureParameters Data Description

The following table explains the content of Procedure Parameters message:

Element	Parent Element	Data Type	Description
ProcedureParameters	DMSMessage		Container. This tag is required
DonationSetupId	ProcedureParameters	Alphanumeric string. 25 characters max.	If using Donation ID as the unique identifier for the procedure setup, populate the DonationSetupId field with the Donation ID. Otherwise, populate the DonationSetupId with the unique identifier for the procedure setup (e.g., 9999999YYYYMM).

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
InstrumentSerialNumber	ProcedureParameters	String. 20 characters max. Must be a serial number of the instrument registered with DXT.	InstrumentSerialNumber is required for PUSH mode configurations. PUSH mode is supported by Aurora and Aurora Xi when configured to have procedures pushed to the instrument. When InstrumentSerialNumber is supplied to DXT, DXT will validate the procedure parameters information, and if it is valid, DXT will immediately forward procedure setup file to the specified instrument. The instrument must be in an available state to accept new parameters; otherwise, the parameters status will be set to an error state and must be resent from DMS. Instruments cannot request (PULL) procedure setup file sent in the PUSH mode.
SiteIdentifier	ProcedureParameters	String. 20 characters max. Must be a site name defined in DMS configurations for the site.	The DMS Site Id associated with the Site Name configured in DXT. Refer to DXT Data Management System Administrator's Guide Chapter 4 for the information about DMS configurations. SiteIdentifier is used to properly associate procedure parameters with the DXT site location for the displaying and reporting purposes.
DXTProcedureParameters	ProcedureParameters		Procedure information complex element. The actual content depends on the instrument type as described in the following sections.

User should verify that DMS Site ID on DXT matches Site ID programmed on the DMS. Incorrect DMS Site ID will prevent DXT from processing remote procedure parameters.

All procedure parameters delivery failures, except operator reject, procedure parameter XML validation failure, or invalid Site ID, are non–recoverable and such parameters must be resent from DMS with the new unique DonationSetupID.

ProcedureParameters DXT Response

Upon successful processing, DXT will respond to procedure parameters with the following XML message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
```


Fresenius Kabi USA, LLC
All rights reserved

```
    <ProcedureParameterStatus>
        <DonationSetupId>DONATION_SETUP_ID</DonationSetupId>
        <ProcedureParametersStatusCode>0</ProcedureParametersStatusCode>
    </ProcedureParameterStatus>
</DMSMessage>
```

DONATION_SETUP_ID in the response will always match the one provided in the request.

The status code in the initial response will always be 0 – Parameters are received by DXT. To monitor updates of Procedure Parameters Status updates, refer to Chapter 3.

Section 2.2 Procedure Parameters for AURORA Instruments

The DMS will use the Procedure Parameters service to send procedure parameters to DXT. This enables the remote procedure setup in push mode if DXT is licensed for this feature.

Refer to Section 2.1 Procedure Parameters for overview of the message. This section defines content and provides examples for messages specific to AURORA instruments.

DXTProcedureParameters XSD Definition

This complex type defines the contents of the Procedure Parameters structure as shown below.

```
<xsd:complexType name="DXTProcedureParameters">
    <xsd:sequence>

        <xsd:element name="ProcedureTypeCode" type="xsd:nonNegativeInteger"
                  minOccurs="0" />

        <xsd:element name="DonationId" minOccurs="0" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:pattern value="[a-zA-Z0-9]*" />
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="DonorId" minOccurs="0" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:pattern value="[a-zA-Z0-9]*" />
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="TargetReplacementFluidVolume" minOccurs="0" maxOccurs="1">
```

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
            <xsd:simpleType>
                <xsd:restriction base="xsd:nonNegativeInteger">
                    <xsd:minInclusive value="0" />
                    <xsd:maxInclusive value="9999" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="TargetPlasmaVolume" minOccurs="0" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:nonNegativeInteger">
                    <xsd:minInclusive value="0" />
                    <xsd:maxInclusive value="9999" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="ACRatio" minOccurs="0" maxOccurs="1">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="5" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="ConfirmationQuestion"
    minOccurs="0" maxOccurs="unbounded">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="100" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>

        <xsd:element name="Questionnaire"
    minOccurs="0" maxOccurs="unbounded">
            <xsd:complexType>
                <xsd:sequence>

                    <xsd:element name="Question" minOccurs="1" maxOccurs="1">
                        <xsd:simpleType>
                            <xsd:restriction base="xsd:string">
                                <xsd:minLength value="1" />
                                <xsd:maxLength value="25" />
                            </xsd:restriction>
                        </xsd:simpleType>
                    </xsd:element>

                    <xsd:element name="Answer" minOccurs="1" maxOccurs="1">
                        <xsd:simpleType>
```


Fresenius Kabi USA, LLC
All rights reserved

```
                        <xsd:restriction base="xsd:string">
                            <xsd:minLength value="1" />
                            <xsd:maxLength value="25" />
                        </xsd:restriction>
                    </xsd:simpleType>
                </xsd:element>
            </xsd:sequence>
        </xsd:complexType>
    </xsd:element>
</xsd:sequence>
</xsd:complexType>
```

DXTProcedureParameters Data Description

The following table explains the content of Procedure Parameters message data for AURORA:

Element	Parent Element	Data Type	Description
DXTProcedureParameters	ProcedureParameters		Container. This tag is required.
ProcedureTypeCode	DXTProcedureParameters	Non–negative integer	Aurora procedure to be configured on the instrument. This field is required. Acceptable values: 11 for Plasma, Saline 12 for Plasma, No Saline Saline protocol must match the instrument configuration for the procedure setup file to be accepted.
DonationId	DXTProcedureParameters	Alphanumeric string. 25 characters max.	Plasma center generated identifier for plasma donation. This field is required.
DonorId	DXTProcedureParameters	Alphanumeric string. 25 characters max.	Plasma center generated unique identifier for the donor. This field is required.
TargetReplacementFluidVolume	DXTProcedureParameters	Integer. 0 – 9999 range.	Target volume of replacement fluid (e.g.: Saline in Plasma procedures) to return to the donor during the procedure, expressed in mL.
TargetPlasmaVolume	DXTProcedureParameters	Integer. 0 – 9999 range.	The target (desired) amount of the Plasma product (plasma with anticoagulant) to be collected by the instrument during the procedure, expressed in mL.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
ACRatio	DXTProcedureParameters	String. 5 characters max.	The ratio of anticoagulant to whole blood used during the procedure. Acceptable values: 6:100 or 8:100
ConfirmationQuestion	DXTProcedureParameters	String. 100 characters max.	An operator question prompt that could result in confirm/reject response.
Questionnaire	DXTProcedureParameters		Container. This tag is required if using questionnaire.
Question	Questionnaire	String. 25 characters max.	Free text question for operator to answer. This field is required if using questionnaire.
Answer	Questionnaire	String. 25 characters max.	An answer to the displayed question for operator to enter. Answer does not appear on the instrument screen and must be entered by operator to confirm the procedure parameters. This field is required if using questionnaire.

When the DMS includes free text questions with procedure parameters, donor's personally identifiable information and applicable laws need to be considered. Instrument will store such information at most for the duration of the procedure and DXT will store such information at most for 24 hours.

Procedure Parameters Example for AURORA Instruments

Following is an example of complete PUSH mode Procedure Parameters exchange for the AURORA instrument with software version 2.0, where instrument is configured to use Donation ID as procedure setup file identifier.

```
POST /fresenius-kabi/dxt/dms/procedure-parameters HTTP/1.1
Date: Tue, 30 Oct 2018 16:00:19 GMT
Content-Type: multipart/mixed; boundary=this_is_fresenius_kabi_http_boundary
Cache-Control: no-cache no-transform
FK-SourceId: 1
FK-Protocol-Version: 3.0.0.0
Host: 5.61.64.115
Content-Length: 1037
Connection: Keep-Alive

--this_is_fresenius_kabi_http_boundary
Content-Type: text/xml; charset=utf-8

<DMSMessage>
    <ProcedureParameters>
        <DonationSetupId>14EILB093031591</DonationSetupId>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
        <InstrumentSerialNumber>PT10005</InstrumentSerialNumber>
        <SiteIdentifier>Site1</SiteIdentifier>
        <DXTProcedureParameters>
            <ProcedureTypeCode>12</ProcedureTypeCode>
            <DonationId>14EILB093031591</DonationId>
            <DonorId>146551591</DonorId>
            <ACRatio>6:100</ACRatio>
            <ConfirmationQuestion>
                Is the donor's birthday May 10, 2018?
            </ConfirmationQuestion>
            <ConfirmationQuestion>
                Is the donor's first name Elizabeth?
            </ConfirmationQuestion>
            <Questionnaire>
                <Question>Last 4 Digits of SSN?</Question>
                <Answer>1234</Answer>
            </Questionnaire>
        </DXTProcedureParameters>
    </ProcedureParameters>
</DMSMessage>

--this_is_fresenius_kabi_http_boundary--
```

Section 2.3 Procedure Parameters for AURORA Xi Instruments

The DMS will use the Procedure Parameters service to send procedure parameters to DXT. This enables the remote procedure setup, both push mode and pull mode, if DXT is licensed for this feature.

Refer to Section 3.1 Procedure Parameters for overview of the message. This section defines content and provides examples for messages specific to AURORA Xi instruments.

DXTProcedureParameters XSD Definition

This complex type defines the contents of the Procedure Parameters structure as shown below.

```
<xsd:simpleType name="DXTInt3">
   <xsd:restriction base="xsd:nonNegativeInteger">
      <xsd:minInclusive value="0" />
      <xsd:maxInclusive value="999" />
   </xsd:restriction>
</xsd:simpleType>

<xsd:simpleType name="DXTFloat2Dot1">
   <xsd:restriction base="xsd:float">
      <xsd:pattern value="[0-9]{1,2}[.][0-9]" />
   </xsd:restriction>
</xsd:simpleType>

<xsd:complexType name="DXTProcedureParameters">
   <xsd:sequence>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
<xsd:element name="ProcedureTypeCode" type="xsd:nonNegativeInteger"
             minOccurs="0" />

<xsd:element name="BloodSampleThisProcedure"
             type="xsd:boolean"
             minOccurs="0" />

<xsd:element name="DonationId" minOccurs="0" maxOccurs="1">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:pattern value="[a-zA-Z0-9]*" />
         <xsd:minLength value="1" />
         <xsd:maxLength value="25" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="DonorId" minOccurs="0" maxOccurs="1">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:pattern value="[a-zA-Z0-9]*" />
         <xsd:minLength value="1" />
         <xsd:maxLength value="25" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="DonorGenderCode" minOccurs="0" maxOccurs="1">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:pattern value="[MF]" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="DonorHeight" minOccurs="0">
   <xsd:complexType>
      <xsd:simpleContent>
         <xsd:extension base="DXTInt3">
            <xsd:attribute name="units" type="xsd:boolean" use="required" />
         </xsd:extension>
      </xsd:simpleContent>
   </xsd:complexType>
</xsd:element>

<xsd:element name="DonorWeight" minOccurs="0">
   <xsd:complexType>
      <xsd:simpleContent>
         <xsd:extension base="DXTInt3">
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
            <xsd:attribute name="units" type="xsd:boolean" use="required" />
         </xsd:extension>
      </xsd:simpleContent>
   </xsd:complexType>
</xsd:element>

<xsd:element name="DonorHematocritOrHemoglobin" minOccurs="0">
   <xsd:complexType>
      <xsd:choice>
         <xsd:element name="DonorHematocrit">
            <xsd:simpleType>
               <xsd:restriction base="xsd:positiveInteger">
                  <xsd:minInclusive value="10" />
                  <xsd:maxInclusive value="99" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="DonorHemoglobin">
            <xsd:complexType>
               <xsd:simpleContent>
                  <xsd:extension base="DXTFloat2Dot1">

                     <xsd:attribute name="units" use="required">
                        <xsd:simpleType>
                           <xsd:restriction
                           base="xsd:nonNegativeInteger">
                              <xsd:minInclusive value="1" />
                              <xsd:maxInclusive value="2" />
                           </xsd:restriction>
                        </xsd:simpleType>
                     </xsd:attribute>

                  </xsd:extension>
               </xsd:simpleContent>
            </xsd:complexType>
         </xsd:element>
      </xsd:choice>
   </xsd:complexType>
</xsd:element>

<xsd:element name="ConfirmationQuestion"
minOccurs="0" maxOccurs=" unbounded ">
   <xsd:simpleType>

      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="100" />
      </xsd:restriction>
   </xsd:simpleType>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
      </xsd:element>

      <xsd:element name="Questionnaire" minOccurs="0" maxOccurs=" unbounded ">
         <xsd:complexType>
            <xsd:sequence>

               <xsd:element name="Question"
               minOccurs="1" maxOccurs="1">
                  <xsd:simpleType>
                     <xsd:restriction base="xsd:string">
                        <xsd:minLength value="1" />
                        <xsd:maxLength value="25" />
                     </xsd:restriction>
                  </xsd:simpleType>
               </xsd:element>

               <xsd:element name="Answer"
               minOccurs="1" maxOccurs="1">
                  <xsd:simpleType>
                     <xsd:restriction base="xsd:string">
                        <xsd:minLength value="1" />
                        <xsd:maxLength value="25" />
                     </xsd:restriction>
                  </xsd:simpleType>
               </xsd:element>

            </xsd:sequence>
         </xsd:complexType>
      </xsd:element>
   </xsd:sequence>
</xsd:complexType>
```

DXTProcedureParameters Data Description

The following table explains the content of Procedure Parameters message data for AURORA Xi:

Element	Parent Element	Data Type	Description
DXTProcedureParameters	ProcedureParameters		Container. This tag is required.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
ProcedureTypeCode	DXTProcedureParameters	Non–negative integer	Aurora Xi procedure to be configured on the instrument. This field is required. Acceptable values are: 11 for Plasma, Saline 12 for Plasma, No Saline Saline protocol must match the instrument configuration for the procedure setup file to be accepted.
BloodSampleThisProcedure	DXTProcedureParameters	Boolean	This parameter instructs the instrument to prompt for a blood sample. If set to true, the instrument will prompt for a blood sample and account for the blood sample volume in blood loss calculations. If set to false, the instrument will not prompt for a blood sample nor account for the blood sample volume in blood loss calculations.
DonationId	DXTProcedureParameters	Alphanumeric string. 25 characters max.	Plasma center generated identifier for plasma donation. This field is required.
DonorId	DXTProcedureParameters	Alphanumeric string. 25 characters max.	Plasma center generated unique identifier for the donor. This field is required.
DonorGenderCode	DXTProcedureParameters	Character	Donor's Gender. Acceptable values are: M (male) or F (female)

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
DonorHeight	DXTProcedureParameters	Integer. 0 – 999 range.	Donor's Height. A "units" attribute must be included in the XML tag. Acceptable values for units are 0 or 1: 0 – Imperial (pounds) 1 – Metric (kilograms) Units must match those configured on the instrument for the procedure setup file to be accepted.
DonorWeight	DXTProcedureParameters	Integer. 0 – 999 range.	Donor's Weight. This field is required. A "units" attribute must be included in the XML tag. Acceptable values for units are 0 or 1: 0 – Imperial (pounds) 1 – Metric (kilograms) Units must match those configured on the instrument for the procedure setup file to be accepted.
DonorHematocritOrHemoglobin	DXTProcedureParameters		Container. This tag is required. Only one of DonorHematocrit or DonorHemoglobin should be populated within this container and must match the instrument configuration for the procedure setup file to be accepted.
DonorHematocrit	DonorHematocritOrHemoglobin	Integer. 36 – 55 range.	Donor's Hematocrit. Either DonorHematocrit or DonorHemoglobin is required. Hematocrit must be within the limits configured on the instrument for the procedure setup file to be accepted.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
DonorHemoglobin	DonorHematocritOrHemoglobin	Float. 11.7 – 17.9 range.	Donor's Hemoglobin. Either DonorHematocrit or DonorHemoglobin is required. Hemoglobin must be within the limits configured on the instrument for the procedure setup file to be accepted. A "units" attribute must be included in the XML tag. Acceptable values for units are 1 or 2: 1 – g/dl 2 – mMol/L Units must match those configured on the instrument for the procedure setup file to be accepted.
TargetPlasmaVolume	DXTProcedureParameters	Integer. 0 – 9999 range.	The target (desired) amount of the Plasma product (plasma with anticoagulant) to be collected by the instrument during the procedure, expressed in mL. Aurora Xi will calculate target product volume based on donor's parameters. The DMS can supply this parameter to overwrite instrument calculated value.
ConfirmationQuestion	DXTProcedureParameters	String. 100 characters max.	An operator question prompt that could result in confirm/reject response.
Questionnaire	DXTProcedureParameters		Container. This tag is required if using questionnaire.
Question	Questionnaire	String. 25 characters max.	Free text question for operator to answer. This field is required if using questionnaire.

Fresenius Kabi USA, LLC
All rights reserved

Element	Parent Element	Data Type	Description
Answer	Questionnaire	String. 25 characters max.	An answer to the displayed question for operator to enter. Answer does not appear on the instrument screen and must be entered by operator to confirm the procedure parameters. This field is required if using questionnaire.

When the DMS includes free text questions with procedure parameters, donor's personally identifiable information and applicable laws need to be considered. Instrument will store such information at most for the duration of the procedure and DXT will store such information at most for 24 hours.

Procedure Parameters Example for AURORA Xi Instruments

Following is an example of complete PULL mode Procedure Parameters exchange for AURORA Xi instrument with software version 1.2, where instrument is configured to use Donation Setup ID as procedure setup file identifier.

```
POST /fresenius-kabi/dxt/dms/procedure-parameters HTTP/1.1
Date: Thu, 01 Nov 2018 14:34:46 GMT
Content-Type: multipart/mixed; boundary=this_is_fresenius_kabi_http_boundary
Cache-Control: no-cache no-transform
FK-SourceId: 1
FK-Protocol-Version: 3.0.0.0
Host: 5.61.64.115
Content-Length: 1290
Expect: 100-continue
--this_is_fresenius_kabi_http_boundary
Content-Type: text/xml; charset=utf-8

<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
   <ProcedureParameters>
      <DonationSetupId>SI14EILB9303159100</DonationSetupId>
      <SiteIdentifier>Site1</SiteIdentifier>
      <DXTProcedureParameters>
         <ProcedureTypeCode>12</ProcedureTypeCode>
         <BloodSampleThisProcedure>true</BloodSampleThisProcedure>
         <DonationId>14EILB093031591a</DonationId>
         <DonorId>146551591</DonorId>
         <DonorGenderCode>M</DonorGenderCode>
         <DonorHeight units="0">72</DonorHeight>
         <DonorWeight units="0">190</DonorWeight>
         <DonorHematocritOrHemoglobin>
            <DonorHematocrit>42</DonorHematocrit>
         </DonorHematocritOrHemoglobin>
         <ConfirmationQuestion>
            Is the donor's birthday May 10, 2018?
         </ConfirmationQuestion>
```

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

```
         <ConfirmationQuestion>
            Is the donor's first name Elizabeth?
         </ConfirmationQuestion>
         <Questionnaire>
            <Question>Last 4 Digits of SSN?</Question>
            <Answer>1234</Answer>
         </Questionnaire>
      </DXTProcedureParameters>
   </ProcedureParameters>
</DMSMessage>
--this_is_fresenius_kabi_http_boundary--
```

Section 2.4 Instrument Status

The DMS will use the Instrument Status service to determine availability of the instruments for the remote procedure setup. This method prevents collisions when the DXT sends parameters in PUSH mode to the instrument that is currently performing procedure or otherwise unable to accept such parameters.

Instrument Status URL

```
GET /fresenius-kabi/dxt/dms/instrument-status/{SiteId} HTTP/1.1
```

{SiteId} is the DMS Site Id associated with the Site Name configured in DXT. Refer to DXT Data Management System Administrator's Guide Chapter 4 for the information about DMS configurations.

Instrument Status DXT Response

Upon successful processing, DXT will respond to instrument status request with the following XML Message:

```
<?xml version="1.0" encoding="utf-8"?>
<DMSMessage>
    <InstrumentStatus>
        ...
    </InstrumentStatus>
</DMSMessage>
```

InstrumentStatus XSD Definition

This complex type defines the contents and structure of the Instrument Status DXT reply XML message.

```
<xsd:complexType name="InstrumentStatus">
   <xsd:sequence>

      <xsd:element name="SiteIdentifier">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:minLength value="1" />
               <xsd:maxLength value="20" />
            </xsd:restriction>
         </xsd:simpleType>
```

Status CURRENT Effective 8/13/2019

All rights reserved

Status CURRENT Effective 8/13/2019

```
</xsd:element>

<xsd:element name="Region">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="50" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="SiteName">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="50" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="Instrument" minOccurs="0"
             maxOccurs="unbounded">
   <xsd:complexType>
      <xsd:sequence>

         <xsd:element name="InstrumentSerialNumber">
            <xsd:simpleType>
               <xsd:restriction base="xsd:string">
                  <xsd:minLength value="1" />
                  <xsd:maxLength value="20" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="InstrumentName">
            <xsd:simpleType>
               <xsd:restriction base="xsd:string">
                  <xsd:minLength value="1" />
                  <xsd:maxLength value="20" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="InstrumentTypeCode"
                      type="xsd:unsignedInt" />

         <xsd:element name="CurrentInstrumentStatusCode"
                      type="xsd:unsignedInt" minOccurs="0"/>

         <xsd:element name="CurrentInstrumentStatusTime"
                      type="xsd:dateTime" minOccurs="0" />
```


Fresenius Kabi USA, LLC
All rights reserved

```
                <xsd:element name="ElapsedCollectionTime"
                             type="xsd:nonNegativeInteger" minOccurs="0" />

                <xsd:element name="RemainingVolume"
                             type="xsd:nonNegativeInteger" minOccurs="0" />

            </xsd:sequence>
          </xsd:complexType>
        </xsd:element>
    </xsd:sequence>
</xsd:complexType>
```

InstrumentStatus Data Description

The following table explains the content of the Instrument Status DXT reply message as applicable to AURORA and AURORA Xi instruments. Refer to the corresponding Data Management supplement for the description of instrument status data for other instruments.

Element	Parent Element	Data Type	Description
InstrumentStatus	DMSMessage		Container. This element is always provided.
SiteIdentifier	InstrumentStatus	String. 20 characters max.	A DMS site name supplied in the request.
Region	InstrumentStatus	String. 50 characters max.	A name of the region configured in DXT.
SiteName	InstrumentStatus	String. 50 characters max.	A name of the site configured in DXT.
Instrument	InstrumentStatus		Container. This element is repeated per instrument configured at the site.
InstrumentSerialNumber	Instrument	String. 20 characters max.	Serial Number of the instrument. This element is always provided.
InstrumentName	Instrument	String. 20 characters max.	Name of the instrument configured in DXT. This element is always provided.
InstrumentTypeCode	Instrument	Integer	Type of the apheresis instrument: 1 – Aurora Xi 3 – Aurora This element is always provided.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
CurrentInstrumentStatusCode	Instrument	Integer	The last known instrument status: 1 – Available. Instrument ready to be setup. 2 – Reserved. Instrument setup has started, but collection has not started. 3 – Collecting. Collection has been started on the instrument. 4 – Post Processing. Collection has been completed on the instrument. Donor ready to be or has been disconnected. 5 – Disconnected. DXT has lost connection with the instrument. 6 – In Service. Instrument is in service mode. This element is omitted if there were no communications with the instrument since DXT Instance has started.
CurrentInstrumentStatusTime	Instrument	Integer	Timestamp of the last status update received from the instrument (instrument status 1, 2, 3, 4, and 6). The time when DXT lost communication with the instrument (instrument status 5). This element is omitted if there were no communications with the instrument since DXT Instance has started.
ElapsedCollectionTime	Instrument	Integer	Time in seconds since the start of the collection. This element is reported only for instruments with status 3 – Collecting.
RemainingVolume	Instrument	Integer	Collection volume remaining calculated as a difference between target and actual volumes. This element is reported only for instruments with status 3 – Collecting.

If multiple DXT Instances are deployed, the Instrument Status request must be sent to the DXT instance hosting the site in the request. See DXT Data Management System Administrator's Guide Installation section for more information about using multiple DXT Instances.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Instrument Status Response Example

Following is an example of Instrument Status DXT reply:

```
FK-Protocol-Version: 3.0.0.0
FK-DestinationId: 1
Content-Length: 1071
Cache-Control: no-cache, no-transform
Content-Type: text/xml; charset=utf-8
Date: Tue, 26 Mar 2019 16:03:39 GMT

<?xml version="1.0"?>
<DMSMessage>
   <InstrumentStatus>
      <SiteIdentifier>DMSSITE1</SiteIdentifier>
      <Region>DXTREG1</Region>
      <SiteName>SITE1</SiteName>
      <Instrument>
         <InstrumentSerialNumber>AU5010</InstrumentSerialNumber>
         <InstrumentName>AU5010</InstrumentName>
         <InstrumentTypeCode>1</InstrumentTypeCode>
         <CurrentInstrumentStatusCode>3</CurrentInstrumentStatusCode>
         <CurrentInstrumentStatusTime>2019-03-
             26T16:03:14.512Z</CurrentInstrumentStatusTime>
         <ElapsedCollectionTime>522</ElapsedCollectionTime>
         <RemainingVolume>543</RemainingVolume>
      </Instrument>
      <Instrument>
         <InstrumentSerialNumber>AU5007</InstrumentSerialNumber>
         <InstrumentName>AU5007</InstrumentName>
         <InstrumentTypeCode>1</InstrumentTypeCode>
         <CurrentInstrumentStatusCode>1</CurrentInstrumentStatusCode>
         <CurrentInstrumentStatusTime>2019-03-
             26T16:03:14.65Z</CurrentInstrumentStatusTime>
      </Instrument>
   </InstrumentStatus>
</DMSMessage>
```

Section 2.5 DXT Server Response

This section lists possible responses that DXT can return to the requests originated from the DMS.

OK Response

DXT will return the following HTTP Response header when the request was processed successfully:

```
HTTP/1.1 200 OK
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
```


Fresenius Kabi USA, LLC
All rights reserved

```
Connection: Close
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0
```

The response may or may not contain XML message, based on the request.

Request Header Validation Error Response

When DXT encounters problems validating the request, the following response will be returned to the sender:

```
HTTP/1.1 400 Bad Request
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

The following table lists Error codes and descriptions for possible errors:

Error Code	Error Description	Notes
51000	FK–SourceId Null or Empty	Sender did not provide “FK–SourceId” header element in the request or the content of “FK–SourceId” header element is empty.
51001	FK–ProtocolVersion Null or Empty	Sender did not provide “FK–ProtocolVersion” header element in the request or the content of “FK–ProtocolVersion” header element is empty.
51002	Date Null or Empty	Sender did not provide “Date” header element in the request or the content of “Date” header element is empty.
51003	Content–Type Null or Empty	Sender did not provide “Content–Type” header element in the request or the content of “Content–Type” header element is empty.
51004	Host Null or Empty	Sender did not provide “Host” header element in the request or the content of “Host” header element is empty.
51005	Cache–Control Null or Empty	Sender did not provide “Cache–Control” header element in the request or the content of “Cache–Control” header element is empty.
51006	Bad FK–ProtocolVersion. Expected: expected version	FK–ProtocolVersion provided in the header is not compatible with DXT version installed.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Error Code	Error Description	Notes
51007	Inconsistent Content–Length Value, actual counted value: actual content length	Content–Length value provided in the header does not match the size of the message received.
51008	Unexpected Content–Type, Expected: Content–Type expected, Received: Content–Type received	Content–Type provided in the header does not match content–type that is expected for the request.
51009	XSD Validation Failure: xml error description	DXT Server has failed to validate the XML message body in the received request. Error description is provided in the "Description" element of the response XML.
51010	Invalid Cache–Control, Expected: Cache–Control expected	Cache–Control provided in the header does not contain "no–cache no–transform".

Request Time Validation Error Response

To ensure timely processing of all inbound messages, the DXT Server requires the message timestamp be within 2 minutes of the DXT Server time, when the message is received.

```
HTTP/1.1 403 Forbidden
Content–Length: 147
Content–Type: text/xml; charset=utf–8
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>51100</Code>
        <Description>Invalid Date value</Description>
    </ErrorResponse>
</DXTMessage>
```

Request Length Validation Error Response

To ensure the integrity of all inbound messages, the DXT Server requires that each request contains Content–Length header element.

When request's Content–Length header element is missing or empty, the following response will be returned to the sender:

```
HTTP/1.1 411 Length Required
Content–Length: 157
Content–Type: text/xml; charset=utf–8
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
```


Fresenius Kabi USA, LLC
All rights reserved

```
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0

 <DXTMessage>
     <ErrorResponse>
         <Code>51200</Code>
         <Description>Content-Length Null or Empty</Description>
         </ErrorResponse>
 </DXTMessage>
```

Request HTTP Version Validation Error Response

When the DXT Server encounters unsupported HTTP version in the request, the following response will be returned to the sender:

```
HTTP/1.1 505 HTTP Version Not Supported
Date: TIMESTAMP
Cache–Control: no–cache no–transform
FK–SourceId: 0
FK–DestinationId: DATA_DESTINATION
FK–Protocol–Version: 3.0.0.0
```

This response does not contain XML message.

Request Processing Error Response

When DXT encounters problems processing the request, the following response will be returned to the sender:

```
HTTP/1.1 500 Internal Server Error
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: 0
FK-DestinationId: DATA_DESTINATION
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>ERROR_CODE</Code>
        <Description>ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

The following table lists Error codes and descriptions for possible errors:

Error Code	Error Description	Notes
50000	Instrument Not Registered	The desired instrument is not registered with DXT or is registered through the DXT Instance other than one the request was sent to.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Error Code	Error Description	Notes
50001	Database Error	DXT has failed to access the database (read or write the data) to process the request.
50009	Remote Procedure Setup Not Licensed	DXT has received Procedure Parameters request, but Remote Procedure Setup feature is not licensed.
50014	Site ID Not Valid	SiteIdentifier element in Procedure Parameters request does not match Site ID configured in DXT for any site location. Or SiteId field in Instrument Status Request URL is not one of the Site Identifier entries configured in Site DMS configuration.
50015	Duplicate Donation Setup ID	DonationSetupId element in Procedure Parameters request contains duplicate/previously received identifier.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Chapter 3 DXT Client

This chapter defines messages that DXT sends to external data management systems.

The destination URL is configurable through DXT User Interface for each data management system DXT communicates with. Refer to DXT Data Management System Administrator's Guide Chapter 4 – DXT Configurations on how to configure DMS interfaces.

For possible error responses from DXT refer to Error Handling section at the end of this chapter.

Section 3.1 Electronic Records for AURORA and AURORA Xi instruments

DXT will use the Electronic Record service to provide the procedure record data to the DMS for records received from AURORA and AURORA Xi instruments, if DXT is licensed for the electronic record feature and DMS is configured on DXT UI.

Electronic Record URL

```
POST {Electronic Record URL}/procedure-record HTTP/1.1
```

DXT will send procedure records to the address configured as "Electronic Record URL" through DXT User Interface.

ElectronicRecord XSD Definition

This complex type defines the contents of the Electronic Record structure. Due to the complexity, the definition is broken into several sub–types, each one presented separately.

```
<xsd:complexType name="ElectronicRecord">
   <xsd:sequence>

      <xsd:element name="ProcedureData">
         <xsd:complexType>
            <xsd:sequence>

               <xsd:element name="ProcedureRecordKey">
                  <xsd:simpleType>
                     <xsd:restriction base="xsd:string">
                        <xsd:minLength value="1" />
                        <xsd:maxLength value="50" />
                     </xsd:restriction>
                  </xsd:simpleType>
               </xsd:element>

               <xsd:element name="InstrumentTypeCode" type="xsd:unsignedInt" />

               <xsd:element name="InstrumentSerialNumber">
                  <xsd:simpleType>
                     <xsd:restriction base="xsd:string">
                        <xsd:minLength value="1" />
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
            <xsd:maxLength value="20" />
         </xsd:restriction>
      </xsd:simpleType>
   </xsd:element>

   <xsd:element name="InstrumentAdministrativeSettingsSignature">
      <xsd:simpleType>
         <xsd:restriction base="xsd:hexBinary">
            <xsd:length value="8" />
         </xsd:restriction>
      </xsd:simpleType>
   </xsd:element>

   <xsd:element name="StartTimeUtc" type="xsd:dateTime" />
   <xsd:element name="EndTimeUtc" type="xsd:dateTime" />

   <xsd:element name="PlasmaRemovedTimeUtc"
                type="xsd:dateTime" minOccurs="0" />

   <xsd:element name="RecordLastUpdateTimeUtc"
                type="xsd:dateTime" />

   <xsd:element name="TimeZone">
      <xsd:simpleType>
         <xsd:restriction base="xsd:string">
            <xsd:minLength value="6" />
            <xsd:maxLength value="6" />
         </xsd:restriction>
      </xsd:simpleType>
   </xsd:element>

   <xsd:element name="ProcedureDeviceControllers">
      <xsd:complexType>
         <xsd:sequence>

            <xsd:element name="ProcedureDeviceController"
                         maxOccurs="unbounded">
               <xsd:complexType>
                  <xsd:sequence>

                     <xsd:element name="Name">
                        <xsd:simpleType>
                           <xsd:restriction base="xsd:string">
                              <xsd:minLength value="1" />
                              <xsd:maxLength value="100" />
                           </xsd:restriction>
                        </xsd:simpleType>
                     </xsd:element>

                     <xsd:element name="Version">
                        <xsd:simpleType>
                           <xsd:restriction base="xsd:string">
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
                                    <xsd:minLength value="1" />
                                    <xsd:maxLength value="20" />
                                 </xsd:restriction>
                              </xsd:simpleType>
                           </xsd:element>
                        </xsd:sequence>
                     </xsd:complexType>
                  </xsd:element>
               </xsd:sequence>
               <xsd:attribute name="count"
                              type="xsd:integer" use="required" />
            </xsd:complexType>
         </xsd:element>

         <xsd:element name="RecordType" type="xsd:unsignedInt" />

         <xsd:element name="ProcedureTypeCode"
                      type="xsd:unsignedInt" minOccurs="0" />

         <xsd:element name="ProcedureType" minOccurs="0">
            <xsd:simpleType>
               <xsd:restriction base="xsd:string">
                  <xsd:minLength value="1" />
                  <xsd:maxLength value="100" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="ProcedureEvents"
         type="DXTProcedureEvents" minOccurs="0"/>

         <xsd:element name="Timespans"
                     type="DXTProcedureTimespans" minOccurs="0"/>

         <xsd:element name="BloodSampleThisProcedure"
                      type="xsd:boolean" minOccurs="0" />

         <xsd:element name="ActiveSWVersion">
            <xsd:simpleType>
               <xsd:restriction base="xsd:string">
                  <xsd:minLength value="1" />
                  <xsd:maxLength value="20" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="ActiveLanguagePackVersion">
            <xsd:simpleType>
               <xsd:restriction base="xsd:string">
                  <xsd:minLength value="1" />
                  <xsd:maxLength value="20" />
               </xsd:restriction>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
                </xsd:simpleType>
             </xsd:element>

             <xsd:choice minOccurs="0">
                <xsd:element name="Donation" type="DXTProcedureDataDonation" />

                <xsd:element name="Demo" type="DXTProcedureDataDonation" />

                <xsd:element name="NonDonation"
                             type="DXTProcedureDataNonDonation" />

             </xsd:choice>

          </xsd:sequence>
       </xsd:complexType>
    </xsd:element>

    <xsd:element name="CompleteStatusCode" type="xsd:unsignedInt" />

    <xsd:element name="CompleteStatusName">
       <xsd:simpleType>
          <xsd:restriction base="xsd:string">
             <xsd:minLength value="1" />
             <xsd:maxLength value="100" />
          </xsd:restriction>
       </xsd:simpleType>
    </xsd:element>
 </xsd:sequence>

 <xsd:attribute name="device">
    <xsd:simpleType>
       <xsd:restriction base="xsd:integer">
          <xsd:enumeration value="1" />
          <xsd:enumeration value="3" />
       </xsd:restriction>
    </xsd:simpleType>
 </xsd:attribute>

</xsd:complexType>
```

ElectronicRecord Data Description

The following table explains the content of ElectronicRecord:

Element	Parent Element	Data Type	Description

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
ProcedureRecord	DMSMessage	ElectronicRecord	Container. This element is always provided. A “device” attribute identifies the device type: 1 – Aurora Xi 3 – Aurora
ProcedureData	ProcedureRecord		Container. This element is always provided.
ProcedureRecordKey	ProcedureData	String. 50 characters max.	Instrument supplied unique identifier of a record. This element is always provided.
InstrumentTypeCode	ProcedureData	Integer	Type of apheresis instrument that performed the procedure: 1 – Aurora Xi 3 – Aurora This element is always provided.
InstrumentSerialNumber	ProcedureData	String. 25 characters max.	Serial Number of the instrument that performed the procedure. This element is always provided.
InstrumentAdministrativeSettingsSignature	ProcedureData	Hexadecimal. 8 bytes.	Unique signature (represented as a 16 hexadecimal) of specific instrument administrative settings configured and used by the instrument during the procedures. This element is always provided.
StartTimeUtc	ProcedureData	dateTime	UTC (Coordinated Universal Time) date and time when the record was created. This time does not necessary associated with actual procedure times. This element is always provided.
EndTimeUtc	ProcedureData	dateTime	UTC (Coordinated Universal Time) date and time when the record was closed. This time does not necessary associated with actual procedure times. This element is always provided.
PlasmaRemovedTimeUtc	ProcedureData	dateTime	UTC (Coordinated Universal Time) date and time of the operator’s confirmation of the plasma container removal from the instrument during the procedure. This element is always provided for Donation records.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
RecordLastUpdateTimeUtc	ProcedureData	dateTime	UTC (Coordinated Universal Time) date and time when DXT received the procedure record from the instrument. This element is always provided.
TimeZone	ProcedureData	String. 6 characters.	Offset value for the local time zone configured on the instrument, represented in format: “+hh:mm” or “–hh:mm”. This element is always provided.
ProcedureDeviceControllers	ProcedureData		Container. This element is always provided. A “count” attribute indicates a number of ProcedureDeviceController child elements in the container.
ProcedureDeviceController	ProcedureDeviceControllers		Container, repeats for each Device Controller.
Name	ProcedureDeviceController	String. 100 characters max.	Name of the device controller. This element is always provided.
Version	ProcedureDeviceController	String. 20 characters max.	Software version identifier of the device controller that was current during the procedure. This element is always provided.
RecordType	ProcedureData	Unsigned integer	Type identifier of the procedure record: 1 – Donation 2 – QC 3 – Non–donation 4 – Service 5 – Demo/Simulation This element is always provided.
ProcedureTypeCode	ProcedureData	Unsigned integer	Type identifier of an apheresis procedure: 11 – Plasma, Saline 12 – Plasma, No Saline This element is always provided for Donation records.
ProcedureType	ProcedureData	String. 100 characters max.	Instrument provided description of a type of an apheresis procedure. This element is always provided for Donation records.

All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
ProcedureEvents	ProcedureData		Container for the instrument supplied events. A “count” attribute indicates the number of ProcedureEvent child elements in the container. This element is omitted if instrument did not supply any events within the record. For the structure and the content, refer to DXTProcedureEvents type definition in this section.
Timespans	ProcedureData		Container for procedure times. A “count” attribute indicates the number of TimeSpan child elements in the container. This element is always provided for Donation records. For the structure and the content, refer to DXTProcedureTimespans type definition in this section.
BloodSampleThisProcedure	ProcedureData	Boolean	Indicator, whether the blood sample was taken from the donor during the procedure: 0 – No 1 – Yes This element is always provided for Donation records by Aurora Xi instruments only.
ActiveSWVersion	ProcedureData	String. 20 characters max	Instrument software version that was current during the procedure. This element is always provided.
ActiveLanguagePackVersion	ProcedureData	String. 20 characters max.	Instrument language pack version that was current during the procedure. This element is always provided.

Fresenius Kabi USA, LLC
All rights reserved

Element	Parent Element	Data Type	Description
Donation	ProcedureData		Container for donation procedure data. This element is always provided for Donation records. For the structure and the content, refer to DXTProcedureDataDonation type definition in this section.
Demo	ProcedureData		Container for demo procedure data. This element is always provided for Demo/Simulation records. For the structure and the content, refer to DXTProcedureDataDonation type definition in this section.
NonDonation	ProcedureData		Container for non–donation record data. This element is always provided for Non–donation records. For the structure and the content, refer to DXTProcedureDataNonDonation type definition in this section.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
CompleteStatusCode	ProcedureRecord	Unsigned integer	The final status of the procedure: 1 – Complete (Nominal procedure, ran to the end without issues) 2 – Terminate with fluid return (Operator terminated the procedure and chose to return fluids to the donor) 3 – Terminate without fluid return (Operator terminated the procedure and chose not to return fluids to the donor) 4 – Terminate shut down (The instrument shut down with non–recoverable alarm during procedure) 5 – Terminate Unknown (The instrument shut down unexpectedly and did not properly close out the record. Content of the record may be unreliable and must be manually confirmed. Not all record fields may be present.) 21 – Pass (QC check procedure has passed on the instrument) 22 – Fail (QC check procedure has failed on the instrument) 23 – Incomplete (QC check procedure has not been completed, i.e., instrument abruptly shut down)
CompleteStatusName	ProcedureRecord	String. 100 characters max.	Instrument provided description of the final status of the procedure. This element is always provided.

DXTProcedureEvents XSD Definition

This complex type defines the contents of the Procedure Events structure as shown below.

```
<xsd:complexType name="DXTProcedureEvents">
   <xsd:sequence>

      <xsd:element name="ProcedureEvent" maxOccurs="unbounded">
         <xsd:complexType>
            <xsd:sequence>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
<xsd:element name="DateTimeUtc" type="xsd:dateTime" />

<xsd:element name="Name">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="200" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="CategoryCode" type="xsd:unsignedInt" />

<xsd:element name="OperatorId" minOccurs="0">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="20" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="AlarmAlertDetail" minOccurs="0">
   <xsd:complexType>
      <xsd:sequence>

         <xsd:element name="AlertAlarmCode">
            <xsd:simpleType>
               <xsd:restriction base="xsd:nonNegativeInteger">
                  <xsd:minInclusive value="0" />
                  <xsd:maxInclusive value="99999" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="AlertAlarmClassCode"
         type="xsd:unsignedInt" />

         <xsd:element name="AlertAlarmClass">
            <xsd:simpleType>
               <xsd:restriction base="xsd:string">
                  <xsd:minLength value="1" />
                  <xsd:maxLength value="50" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="AlarmData1" type="xsd:int" />
         <xsd:element name="AlarmData2" type="xsd:int" />
      </xsd:sequence>
   </xsd:complexType>
</xsd:element>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
                <xsd:element name="QCEventDetail" minOccurs="0">
                   <xsd:complexType>
                      <xsd:sequence>

                         <xsd:element name="QCEventDetails">
                            <xsd:simpleType>
                               <xsd:restriction base="xsd:string">
                                  <xsd:minLength value="1" />
                                  <xsd:maxLength value="100" />
                               </xsd:restriction>
                            </xsd:simpleType>
                         </xsd:element>

                         <xsd:element name="QCEventResult">
                            <xsd:simpleType>
                               <xsd:restriction base="xsd:string">
                                  <xsd:minLength value="1" />
                                  <xsd:maxLength value="50" />
                               </xsd:restriction>
                            </xsd:simpleType>
                         </xsd:element>

                         <xsd:element name="QCEventResultCode" type="xsd:boolean" />

                         <xsd:element name="QCEventScaleType">
                            <xsd:simpleType>
                               <xsd:restriction base="xsd:string">
                                  <xsd:minLength value="1" />
                                  <xsd:maxLength value="50" />
                               </xsd:restriction>
                            </xsd:simpleType>
                         </xsd:element>

                         <xsd:element name="QCEventScaleTypeCode"
                                      type="xsd:unsignedInt" />
                      </xsd:sequence>
                   </xsd:complexType>
                </xsd:element>
             </xsd:sequence>
          </xsd:complexType>
       </xsd:element>
    </xsd:sequence>

    <xsd:attribute name="count" type="xsd:integer" use="required" />

</xsd:complexType>
```

DXTProcedureEvents Data Description

The following table explains the content of DXTProcedureEvents:

Fresenius Kabi USA, LLC
All rights reserved

Element	Parent Element	Data Type	Description
ProcedureEvents	ProcedureData		Container for the instrument supplied events. A “count” attribute indicates the number of ProcedureEvent child elements in the container. This element is omitted if instrument did not supply any events within the record.
ProcedureEvent	ProcedureEvents		Container, repeats for each instrument supplied event. All events appear in the sequence in the chronological order.
DateTimeUtc	ProcedureEvent	dateTime	UTC (Coordinated Universal Time) date and time of an event occurred during the procedure and reported by the instrument. This element is always provided.
Name	ProcedureEvent	String. 200 characters max.	Text description of an event reported by the instrument. This element is always provided. Note that all event descriptions are in the instrument’s language. The exact descriptions of the events will change if instrument’s language is changed.

Status CURRENT Effective 8/13/2019

All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
CategoryCode	ProcedureEvent	Unsigned integer	Event type identifier: 1 – Note Added (Operator entered note during the procedure. The exact note will appear as part of event's description. See Custom Events notes below.) 2 – Alarm/Alert (Instrument triggered alarm/alert events) 3 – Instrument Events (e.g., switch to battery power, etc.) 4 – Operator Tasks (Operator interactions with the instrument, except operator added notes.) 5 – QC Weigh Scale Check 6 – QC Weigh Scale Calibration 7 – Service Maintenance (Service and maintenance events, except for weigh scale activities.) 8 – Alarm/Alert Clear This element is always provided.
OperatorId	ProcedureEvent	String. 20 characters max.	The ID of the Operator signed into the instrument at the time of the event occurred.
AlarmAlertDetail	ProcedureEvent		Container for the details about alarm/alert events. This element is only provided for Alarm/Alert events.
AlertAlarmCode	AlarmAlertDetail	Integer. 0–99999 range.	The alarm/alert code supplied by the instrument. Refer to the specific instrument's label copy for explanation of these codes. This element is always provided.
AlertAlarmClassCode	AlarmAlertDetail	Unsigned integer	Severity of the alarm/alert event: 1 – High 2 – Medium 3 – Low 4 – Alert This element is always provided.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
AlertAlarmClass	AlarmAlertDetail	String. 50 characters max.	Instrument provided description of a type of an alarm/alert event severity. This element is always provided.
AlarmData1	AlarmAlertDetail	Integer	Additional data provided by the instrument about an alarm/alert event. This information is event specific. Refer to the specific instrument's label copy for explanation of an additional data. This element is always provided.
AlarmData2	AlarmAlertDetail	Integer	Additional data provided by the instrument about an alarm/alert event. This information is event specific. Refer to the specific instrument's label copy for explanation of an additional data. This element is always provided.
QCEventDetail	ProcedureEvent		Container for the details about QC events. This element is only provided for QC Weigh Scale events by Aurora instruments. This element is only provided for QC Weigh Scale and QC Weigh Scale Calibration events by Aurora Xi instruments.
QCEventDetails	QCEventDetail	String. 100 characters max.	Instrument supplied description of the QC Event. This element is always provided.
QCEventResult	QCEventDetail	String. 50 characters max.	Instrument supplied description of the QC Event. This element is always provided.
QCEventResultCode	QCEventDetail	Boolean	QC Event result: 0 – Fail 1 – Pass This element is always provided.
QCEventScaleType	QCEventDetail	String. 50 characters max.	Instrument supplied description of the scale checked/calibrated during QC event. This element is always provided.
QCEventScaleTypeCode	QCEventDetail	Unsigned integer	Scale checked/calibrated during QC event: 1 – AC (Aurora Xi instruments only) 2 – Reservoir (Aurora Xi instruments only) 3 – Plasma This element is always provided.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Custom Events

The operator can record additional events/notes during the procedure by manually entering them at the instrument or by scanning appropriate bar code. For bar code scanning, GS1–128 symbology is recommended and Application Identifier “96” must be used to scan bar code in as a procedure note. The instrument allows up to 30 characters in total for each Note event.

For example, if this bar code is scanned,



the following event will be present in the procedure’s XML:

```
<ProcedureEvent>
   <DateTimeUtc>2018-11-28T19:54:40</DateTimeUtc>
   <Name>Note Added: Operator's Note</Name>
   <CategoryCode>1</CategoryCode>
   <OperatorId>Operator</OperatorId>
</ProcedureEvent>
```

DXTProcedureTimespans XSD Definition

This complex type defines the contents of the DXTProcedureTimespans structure as shown below.

```
<xsd:complexType name="DXTProcedureTimespans">
   <xsd:sequence>

      <xsd:element name="Timespan" maxOccurs="unbounded">
         <xsd:complexType>
            <xsd:sequence>

               <xsd:element name="TypeCode" type="xsd:unsignedInt" />

               <xsd:element name="StartTime" type="xsd:dateTime" />

               <xsd:element name="EndTime" type="xsd:dateTime" minOccurs="0" />

               <xsd:element name="Duration" minOccurs="0">
                  <xsd:simpleType>
                     <xsd:restriction base="xsd:nonNegativeInteger">
                        <xsd:minInclusive value="0" />
                        <xsd:maxInclusive value="99999" />
                     </xsd:restriction>
                  </xsd:simpleType>
               </xsd:element>
```


Fresenius Kabi USA, LLC
All rights reserved

```
            </xsd:sequence>
          </xsd:complexType>
        </xsd:element>
      </xsd:sequence>

      <xsd:attribute name="count" type="xsd:integer" />
</xsd:complexType>
```

DXTProcedureTimespans Data Description

The following table explains the content of DXTProcedureTimespans:

Element	Parent Element	Data Type	Description
Timespans	ProcedureData		Container for procedure times. A “count” attribute indicates the number of TimeSpan child elements in the container. This element is always provided for Donation records.
TimeSpan	Timespans		Container for the specific timespan of the apheresis procedure recorded by the instrument.
TypeCode	TimeSpan	Unsigned integer	Type of the timespan: 1 – Venipuncture (Donor connected to the instrument) 2 – Collection (Actual collection of plasma product by the instrument)
StartTime	TimeSpan	dateTime	UTC (Coordinated Universal Time) date and time of the timespan start event: Venipuncture: Operator confirmation of the needle stick. Collection: Start of the first draw cycle.
EndTime	TimeSpan	dateTime	UTC (Coordinated Universal Time) date and time of the timespan end event: Venipuncture: Operator confirmation of the needle removal. Collection: End time of the last return cycle. This element may be omitted if end event was not recorded due to abnormal termination.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Element	Parent Element	Data Type	Description
Duration	TimeSpan	Integer	The duration in seconds between timespan's StartTime and EndTime. This element is provided if both start and end times were recorded.

DXTProcedureDataDonation XSD Definition

This complex type defines the contents of the DXTProcedureDataDonation structure as shown below.

```
<xsd:complexType name="DXTProcedureDataDonation">
   <xsd:sequence>

      <xsd:element name="DonorGenderCode" minOccurs="0">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:pattern value="[MF]" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="DonorHeight" minOccurs="0">
         <xsd:complexType>
            <xsd:simpleContent>
               <xsd:extension base="xsd:integer">
                  <xsd:attribute name="units" type="xsd:boolean" use="required" />
               </xsd:extension>
            </xsd:simpleContent>
         </xsd:complexType>
      </xsd:element>

      <xsd:element name="DonorWeight" minOccurs="0">
         <xsd:complexType>
            <xsd:simpleContent>
               <xsd:extension base="xsd:integer">
                  <xsd:attribute name="units" type="xsd:boolean" use="required" />
               </xsd:extension>
            </xsd:simpleContent>
         </xsd:complexType>
      </xsd:element>

      <xsd:element name="DonorHematocritOrHemoglobin" minOccurs="0">
         <xsd:complexType>
            <xsd:choice>

               <xsd:element name="DonorHematocrit" >
                  <xsd:simpleType >
                     <xsd:restriction base="xsd:positiveInteger">
                        <xsd:minInclusive value="10" />
                        <xsd:maxInclusive value="99" />
                     </xsd:restriction>
                  </xsd:simpleType>
```

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
            </xsd:element>

            <xsd:element name="DonorHemoglobin">
               <xsd:complexType>
                  <xsd:simpleContent>
                     <xsd:extension base="xsd:float">
                        <xsd:attribute name="units" use="required" />
                     </xsd:extension>
                  </xsd:simpleContent>
               </xsd:complexType>
            </xsd:element>

         </xsd:choice>
      </xsd:complexType>
   </xsd:element>

   <xsd:element name="TargetReplacementFluidVolume" type="xsd:nonNegativeInteger"
                minOccurs="0" />

   <xsd:element name="DonationSetupId" minOccurs="0">
      <xsd:simpleType>
         <xsd:restriction base="xsd:string">
            <xsd:minLength value="1" />
            <xsd:maxLength value="25" />
         </xsd:restriction>
      </xsd:simpleType>
   </xsd:element>

   <xsd:element name="DonationId" minOccurs="0">
      <xsd:simpleType>
         <xsd:restriction base="xsd:string">
            <xsd:minLength value="1" />
            <xsd:maxLength value="25" />
         </xsd:restriction>
      </xsd:simpleType>
   </xsd:element>

   <xsd:element name="DonorId" minOccurs="0">
      <xsd:simpleType>
         <xsd:restriction base="xsd:string">
            <xsd:minLength value="1" />
            <xsd:maxLength value="25" />
         </xsd:restriction>
      </xsd:simpleType>
   </xsd:element>

   <xsd:element name="OperatorStartId" minOccurs="0">
      <xsd:simpleType>
         <xsd:restriction base="xsd:string">
            <xsd:minLength value="1" />
            <xsd:maxLength value="20" />
         </xsd:restriction>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
   </xsd:simpleType>
</xsd:element>

<xsd:element name="OperatorEndId" minOccurs="0">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="20" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="PhlebotomistStartId" minOccurs="0">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="20" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="PhlebotomistEndId" minOccurs="0" >
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="20" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="WBVolumeProcessed" type="xsd:nonNegativeInteger"
             minOccurs="0" />

<xsd:element name="TotalCycles" type="xsd:integer"minOccurs="0" />

<xsd:element name="ACUsed" type="xsd:nonNegativeInteger" minOccurs="0" />

<xsd:element name="ACRatio" minOccurs="0">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="6" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="ACSolution" minOccurs="0">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="15" />
      </xsd:restriction>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="SalineUsed" type="xsd:nonNegativeInteger" minOccurs="0" />

      <xsd:element name="RBCKitVolume" type="xsd:integer" minOccurs="0" />

      <xsd:element name="ProcedureSoftGoods"
                   type="DXTProcedureSoftGoods"
                   minOccurs="0" />

      <xsd:element name="ProcedureProducts"
                   type="DXTProcedureProducts"
                   minOccurs="0" />

   </xsd:sequence>
</xsd:complexType>
```

DXTProcedureDataNonDonation XSD Definition

This complex type defines the contents of the DXTProcedureDataNonDonation structure as shown below.

```
<xsd:complexType name="DXTProcedureDataNonDonation">
   <xsd:sequence>

      <xsd:element name="DonorGenderCode" minOccurs="0">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:pattern value="[MF]" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="DonorHeight" minOccurs="0">
         <xsd:complexType>
            <xsd:simpleContent>
               <xsd:extension base="xsd:integer">
                  <xsd:attribute name="units" type="xsd:boolean" use="required" />
               </xsd:extension>
            </xsd:simpleContent>
         </xsd:complexType>
      </xsd:element>

      <xsd:element name="DonorWeight" minOccurs="0">
         <xsd:complexType>
            <xsd:simpleContent>
               <xsd:extension base="xsd:integer">
                  <xsd:attribute name="units" type="xsd:boolean" use="required" />
               </xsd:extension>
            </xsd:simpleContent>
         </xsd:complexType>
      </xsd:element>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
<xsd:element name="DonorHematocritOrHemoglobin" minOccurs="0">
   <xsd:complexType>
      <xsd:choice>
         <xsd:element name="DonorHematocrit">
            <xsd:simpleType>
               <xsd:restriction base="xsd:positiveInteger">
                  <xsd:minInclusive value="10" />
                  <xsd:maxInclusive value="99" />
               </xsd:restriction>
            </xsd:simpleType>
         </xsd:element>

         <xsd:element name="DonorHemoglobin">
            <xsd:complexType>
               <xsd:simpleContent>
                  <xsd:extension base="xsd:float">
                     <xsd:attribute name="units"
                                    type="xsd:integer"
                                    use="required" />
                  </xsd:extension>
               </xsd:simpleContent>
            </xsd:complexType>
         </xsd:element>
      </xsd:choice>
   </xsd:complexType>
</xsd:element>

<xsd:element name="TargetReplacementFluidVolume"
             type="xsd:nonNegativeInteger"
             minOccurs="0" />

<xsd:element name="DonationSetupId" minOccurs="0">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="25" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="DonationId" minOccurs="0">
   <xsd:simpleType>
      <xsd:restriction base="xsd:string">
         <xsd:minLength value="1" />
         <xsd:maxLength value="25" />
      </xsd:restriction>
   </xsd:simpleType>
</xsd:element>

<xsd:element name="DonorId" minOccurs="0">
   <xsd:simpleType>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
            <xsd:restriction base="xsd:string">
               <xsd:minLength value="1" />
               <xsd:maxLength value="25" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="OperatorStartId" minOccurs="0">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:minLength value="1" />
               <xsd:maxLength value="20" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="OperatorEndId" minOccurs="0">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:minLength value="1" />
               <xsd:maxLength value="20" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="ProcedureSoftGoods"
                   type="DXTProcedureSoftGoods"
                   minOccurs="0" />

   </xsd:sequence>

</xsd:complexType>
```

DXTProcedureDataDonation and DXTProcedureDataNonDonation Data Description

DXT will forward non–donation and demo records only if appropriate record type is selected during DMS configuration. Refer to DXT Data Management System Administrator's Guide Chapter 4 – DXT Configurations on how to configure DMS interfaces.

The following table explains the content of DXTProcedureDataDonation:

Element	Parent Element	Data Type	Description
Donation, NonDonation or Demo	ProcedureData		Container for donation, non–donation, or demo procedure data. This element is always provided for Donation and Demo records.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
DonorGenderCode	Donation, NonDonation or Demo	Character	Donor's Gender: M – Male F – Female This element is provided by Aurora Xi instruments only.
DonorHeight	Donation, NonDonation or Demo	Integer	Donor's height as entered by the operator at the instrument or sent by the DMS in the remote procedure setup. A "units" attribute indicates units of measurement: 0 – Imperial (inches) 1 – Metric (centimeters) This element is provided by Aurora Xi instruments
DonorWeight	Donation, NonDonation or Demo	Integer	Donor's weight as entered by the operator at the instrument or sent by the DMS in the remote procedure setup. A "units" attribute indicates units of measurement: 0 – Imperial (inches) 1 – Metric (centimeters) This element is provided by Aurora Xi instruments.
DonorHematocritOrHemoglobin	Donation, NonDonation or Demo		Container for donor's Hematocrit or Hemoglobin data. This element is provided by Aurora Xi instruments only.
DonorHematocrit	DonorHematocritOrHemoglobin	Integer. 10 – 99 range.	Donor's hematocrit as entered by the operator at the instrument or sent by the DMS in the remote procedure setup. Either DonorHematocrit or DonorHemoglobin is always provided.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
DonorHemoglobin	DonorHematocritOrHemoglobin	Float	Donor's hemoglobin as entered by the operator at the instrument or sent by the DMS in the remote procedure setup. A "units" attribute indicates units of measurement: 1 – g/dL 2 – mMol/L Either DonorHematocrit or DonorHemoglobin is always provided.
TargetReplacementFluidVolume	Donation, NonDonation or Demo	Integer	The target volume, in milliliters, for the replacement fluid (e.g., Saline) to be returned to donor during the procedure.
DonationSetupId	Donation, NonDonation or Demo	String. 25 characters max.	Identifier for the procedure parameters that was created and sent by the DMS as a part of procedure setup file or entered by the operator at the instrument.
DonationId	Donation, NonDonation or Demo	String. 25 characters max.	Procedure identifier as entered by the operator at the instrument or sent by the DMS in the procedure setup file.
DonorId	Donation, NonDonation or Demo	String. 25 characters max.	Donor identifier as entered by the operator at the instrument or sent by the DMS in the procedure setup file.
OperatorStartId	Donation, NonDonation or Demo	String. 20 characters max.	ID of the operator logged in at the instrument at the time of the start of the procedure.
OperatorEndId	Donation, NonDonation or Demo	String. 20 characters max.	ID of the operator logged in at the instrument at the time of the confirmation of bottle removal.
PhlebotomistStartId	Donation or Demo	String. 20 characters max.	ID of the operator logged in at the instrument at the time of the confirmation of venipuncture.
PhlebotomistEndId	Donation or Demo	String. 20 characters max.	ID of the operator logged in at the instrument at the time of the confirmation of needle removal.
WBVolumeProcessed	Donation or Demo	Integer	Volume of whole blood, in milliliters, processed from the donor during collection.
TotalCycles	Donation or Demo	Integer	Total number of draw–return cycles completed during procedure.

All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
ACUsed	Donation or Demo	Integer	The total volume of anticoagulant, in milliliters, used during the procedure.
ACRatio	Donation or Demo	String. 6 characters max.	The ratio of anticoagulant to whole blood used during the procedure as entered by the operator at the instrument or sent by DMS in the remote procedure setup. Ratio supported by Aurora Xi instruments: “1:6”. Ratio supported by Aurora instruments: “6:100”, “8:100”, “6%”, “8%”.
ACSolution	Donation or Demo	String. 15 characters max.	The name/description of anticoagulant used during the procedure. This element is provided by Aurora Xi instruments only.
SalineUsed	Donation or Demo	Integer	Actual volume of saline, in milliliters, used during the procedure.
RBCKitVolume	Donation or Demo	Integer	RBC volume, in milliliters, remaining within kit after the procedure. This element is provided by Aurora Xi instruments only.
ProcedureSoftGoods	Donation, NonDonation or Demo		Container for reporting soft goods/disposables used during the procedure. A “count” attribute indicates the number of ProcedureSoftGood child elements in the container. For the structure and the content, refer to DXTProcedureSoftGoods type definition in this section.
ProcedureProducts	Donation or Demo		Container for reporting products collected during the procedure. A “count” attribute indicates a number of ProcedureProduct child elements in the container. For the structure and the content, refer to DXTProcedureProducts type definition in this section.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

DXTProcedureSoftGoods XSD Definition

This complex type defines the contents of the DXTProcedureSoftGoods structure as shown below.

```
<xsd:complexType name="DXTProcedureSoftGoods">
   <xsd:sequence>

      <xsd:element name="ProcedureSoftGoodTypeCode" type="xsd:unsignedInt" />

      <xsd:element name="ProcedureSoftGoodType">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:minLength value="1" />
               <xsd:maxLength value="50" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="REF" minOccurs="0">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:minLength value="1" />
               <xsd:maxLength value="2030" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="Lot" minOccurs="0">
         <xsd:simpleType>
            <xsd:restriction base="xsd:string">
               <xsd:minLength value="1" />
               <xsd:maxLength value="30" />
            </xsd:restriction>
         </xsd:simpleType>
      </xsd:element>

      <xsd:element name="Expiration" type="xsd:date" minOccurs="0" />

   </xsd:sequence>
</xsd:complexType>
```

DXTProcedureSoftGoods Data Description

The following table explains the content of DXTProcedureSoftGoods:

Element	Parent Element	Data Type	Description

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
ProcedureSoftGoods	Donation, NonDonation or Demo		Container for reporting soft goods /disposables used during the procedure. A “count” attribute indicates the number of ProcedureSoftGood child elements in the container.
ProcedureSoftGood	ProcedureSoftGoods		Container for reporting each individual soft good/disposable set used during the procedure. All soft goods/disposable sets scanned or entered on the instrument will be reported. In the case multiple soft goods/disposable sets of the same type are present, to determine which one was used during the procedure, procedure events will need to be analyzed.
ProcedureSoftGoodTypeCode	ProcedureSoftGood	Unsigned integer	Soft good/disposable set type: 1 – Disposable Set 2 – Anticoagulant 4 – Saline 5 – Needle Set 6 – Collection Container 7 – Miscellaneous (Aurora Xi instruments only) This element is always provided.
ProcedureSoftGoodType	ProcedureSoftGood	String. 50 characters max.	Text description of a soft goods type as reported by the instrument. This element is always provided.
REF	ProcedureSoftGood	String. 30 characters max.	The Fresenius Kabi product code of the soft good/disposable set used during the procedure.
Lot	ProcedureSoftGood	String. 20 characters max.	The lot number of the soft good/disposable set used during the procedure.
Expiration	ProcedureSoftGood	Date	The expiration date of the soft good/disposable set used during the procedure.

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

DXTProcedureProducts XSD Definition

This complex type defines the contents of the DXTProcedureProducts structure as shown below.

```
<xsd:complexType name="DXTProcedureProducts">
   <xsd:sequence>

      <xsd:element name="ProcedureProduct" maxOccurs="unbounded">
         <xsd:complexType>
            <xsd:sequence>

               <xsd:element name="ProductExceptions" minOccurs="0">
                  <xsd:complexType>
                     <xsd:sequence>

                        <xsd:element name="ProductException" maxOccurs="unbounded">
                           <xsd:complexType>
                              <xsd:sequence>

                                 <xsd:element name="Exception">
                                    <xsd:simpleType>
                                       <xsd:restriction base="xsd:string">
                                          <xsd:minLength value="1" />
                                          <xsd:maxLength value="50" />
                                       </xsd:restriction>
                                    </xsd:simpleType>
                                 </xsd:element>

                                 <xsd:element name="Code" type="xsd:unsignedInt" />

                              </xsd:sequence>
                           </xsd:complexType>
                        </xsd:element>

                     </xsd:sequence>
                     <xsd:attribute name="count" type="xsd:integer" use="required" />
                  </xsd:complexType>
               </xsd:element>

               <xsd:element name="ProductSubComponents">
                  <xsd:complexType>
                     <xsd:sequence>

                        <xsd:element name="ProductSubComponent"
                                     maxOccurs="unbounded">
                           <xsd:complexType>
                              <xsd:sequence>

                                 <xsd:element name="Code" type="xsd:unsignedInt" />
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
                                    <xsd:element name="Type">
                                       <xsd:simpleType>
                                          <xsd:restriction base="xsd:string">
                                             <xsd:minLength value="1" />
                                             <xsd:maxLength value="50" />
                                          </xsd:restriction>
                                       </xsd:simpleType>
                                    </xsd:element>

                                    <xsd:element name="IsPrimary" type="xsd:boolean" />

                                    <xsd:element name="VolumeData">
                                       <xsd:complexType>
                                          <xsd:sequence>

                                             <xsd:element
                                                 name="TargetVolume"
                                                 type="xsd:nonNegativeInteger" />
                                             <xsd:element
                                                 name="Volume"
                                                 type="xsd:integer" />
                                          </xsd:sequence>
                                       </xsd:complexType>
                                    </xsd:element>
                                 </xsd:sequence>
                              </xsd:complexType>
                           </xsd:element>
                        </xsd:sequence>

                        <xsd:attribute name="count" type="xsd:integer" use="required" />
                     </xsd:complexType>
                  </xsd:element>

               </xsd:sequence>
            </xsd:complexType>
         </xsd:element>

      </xsd:sequence>
      <xsd:attribute name="count" type="xsd:integer" use="required" />
   </xsd:complexType>
```

DXTProcedureProducts Data Description

The following table explains the content of DXTProcedureProducts:

Element	Parent Element	Data Type	Description

Fresenius Kabi USA, LLC
All rights reserved

Element	Parent Element	Data Type	Description
ProcedureProducts	Donation or Demo		Container for reporting products collected during the procedure. A “count” attribute indicates the number of ProcedureProduct child elements in the container. This element is always provided for Donation records.
ProcedureProduct	ProcedureProducts		Container for each product collected during the procedure. Only one product is reported for Aurora and Aurora Xi procedures.
ProductExceptions	ProcedureProduct		Container for reporting product related exceptions. A “count” attribute indicates the number of ProductException child elements in the container.
ProductException	ProductExceptions		Container for each exception associated with the product.
Exception	ProductException	String. 50 characters max.	Text description of an exception as reported by the instrument. This element is always provided.
Code	ProductException	Unsigned integer	Exception type: 11 – Verify Product 12 – Verify Red Cell Loss (Aurora Xi instruments only) This element is always provided.
ProductSubComponents	ProcedureProduct		Container for reporting product components. A “count” attribute indicates the number of ProductSubComponent child elements in the container. This element is always provided.
ProductSubComponent	ProductSubComponents		Container for reporting each product component. At least one ProductSubComponent is provided per product.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

Element	Parent Element	Data Type	Description
Code	ProductSubComponent	Unsigned integer	Type of the product component: 1 – Anticoagulated Plasma (Total plasma product. Primary component.) 3 – Anticoagulant (Aurora Xi instruments only) 8 – Plasma Volume (Plasma without AC; Aurora Xi instruments only)
Type	ProductSubComponent	String. 50 characters max.	Text description of the component type as reported by the instrument. This element is always provided.
IsPrimary	ProductSubComponent	Boolean	Indicator of whether the component is primary for the product: 0 – Secondary 1 – Primary
VolumeData	ProductSubComponent		Container for target and actual component volumes.
TargetVolume	VolumeData	Integer	The target amount, in milliliters, of the component in the final product to be collected. TargetVolume for Anticoagulant (Code: 3) is always reported as zero.
Volume	VolumeData	Integer	The actual amount, in milliliters, of the component in the final product. For the primary component, this is final volume of the collected product. Note, this amount may be negative if operator removes container before instrument prompts for container removal.

Electronic Record Example for Donation Record

Following is an example of complete Donation Procedure Record exchange between DXT and the DMS, for Aurora instrument with software version 2.0.

```
POST /procedure-record HTTP/1.1
Connection: keep-alive,Keep-Alive
Date: Wed, 27 Mar 2019 03:59:01 GMT
FK-SourceId: 0
FK-Protocol-Version: 3.0.0.0
Cache-Control: no-transform, no-cache
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
Content-Type: text/xml; charset=utf-8
Host: 172.16.10.23:8060
Content-Length: 30026

<?xml version="1.0"?>
<DMSMessage>
   <ProcedureRecord device="3">
      <ProcedureData>
         <ProcedureRecordKey>20190322063856_BC09933_PR</ProcedureRecordKey>
         <InstrumentTypeCode>3</InstrumentTypeCode>
         <InstrumentSerialNumber>BC09933</InstrumentSerialNumber>
         <InstrumentAdministrativeSettingsSignature>1446836972A59C2B
         </InstrumentAdministrativeSettingsSignature>
         <StartTimeUtc>2019-03-22T11:38:56</StartTimeUtc>
         <EndTimeUtc>2019-03-22T15:55:06</EndTimeUtc>
         <PlasmaRemovedTimeUtc>2019-03-22T15:55:06</PlasmaRemovedTimeUtc>
         <RecordLastUpdateTimeUtc>2019-03-22T15:55:09</RecordLastUpdateTimeUtc>
         <TimeZone>-05:00</TimeZone>
         <ProcedureDeviceControllers count="3">
            <ProcedureDeviceController>
               <Name>Hardware Controller SW Version</Name>
               <Version>10</Version>
            </ProcedureDeviceController>
            <ProcedureDeviceController>
               <Name>MPU SW Version</Name>
               <Version>2.0.18.109</Version>
            </ProcedureDeviceController>
            <ProcedureDeviceController>
               <Name>Safety Controller SW Version</Name>
               <Version>1</Version>
            </ProcedureDeviceController>
         </ProcedureDeviceControllers>
         <RecordType>1</RecordType>
         <ProcedureTypeCode>11</ProcedureTypeCode>
         <ProcedureType>Plasma Collection, Saline</ProcedureType>
         <ProcedureEvents count="92">
            <ProcedureEvent>
               <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
               <Name>Changed: Target Saline Infusion Volume->500</Name>
               <CategoryCode>4</CategoryCode>
               <OperatorId>FRED</OperatorId>
            </ProcedureEvent>
            <ProcedureEvent>
               <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
               <Name>Changed: Target Return Rate->150</Name>
               <CategoryCode>4</CategoryCode>
               <OperatorId>FRED</OperatorId>
            </ProcedureEvent>
            <ProcedureEvent>
               <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
               <Name>Changed: Procedure ID-></Name>
               <CategoryCode>4</CategoryCode>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
      <Name>Changed: Donation Setup ID-></Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
      <Name>Changed: Donor ID-></Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
      <Name>Changed: Cuff Pressure Draw Setting->52.00000</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
      <Name>Changed: Target Draw Rate->120</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T11:38:56</DateTimeUtc>
      <Name>Changed: Squeeze Threshold->-110</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T11:43:56</DateTimeUtc>
      <Name>Changed: Operator ID-></Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:36:17</DateTimeUtc>
      <Name>Changed: Operator ID->MARY</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARY</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:36:20</DateTimeUtc>
      <Name>Resume From Stand by Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARY</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:40:50</DateTimeUtc>
      <Name>Install Disposable Set Task Done</Name>
```


All rights reserved

Status CURRENT Effective 8/13/2019


```
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARY</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:45:50</DateTimeUtc>
      <Name>Changed: Operator ID-></Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:48:39</DateTimeUtc>
      <Name>Changed: Operator ID->MARK</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:48:42</DateTimeUtc>
      <Name>Connect Solutions Task Done</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:48:54</DateTimeUtc>
      <Name>Changed: Procedure ID->AT3053921</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:51:26</DateTimeUtc>
      <Name>Changed: Target Plasma Collection Volume->880</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:56:25</DateTimeUtc>
      <Name>Check Plasma Container Connection Task Done</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:56:27</DateTimeUtc>
      <Name>Venipuncture Task Done, Ready to Start</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:56:29</DateTimeUtc>
      <Name>Begin Collection</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T13:56:38</DateTimeUtc>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <Name>Changed: Target Return Rate->120</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>MARK</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:04:42</DateTimeUtc>
      <Name>Changed: Operator ID-></Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:05:07</DateTimeUtc>
      <Name>Auto Draw Rate Decrease</Name>
      <CategoryCode>3</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:05:07</DateTimeUtc>
      <Name>Changed: Target Draw Rate->110</Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:08:49</DateTimeUtc>
      <Name>3508(0,0) No Blood Flow</Name>
      <CategoryCode>2</CategoryCode>
      <AlarmAlertDetail>
         <AlertAlarmCode>3508</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:08:50</DateTimeUtc>
      <Name>Changed: Target Draw Rate->100</Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:09:02</DateTimeUtc>
      <Name>Changed: Operator ID->FRED</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:09:03</DateTimeUtc>
      <Name>Cleared AlertAlarm.3508</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:09:03</DateTimeUtc>
      <Name>Acknowledged AlertAlarm.</Name>
      <CategoryCode>4</CategoryCode>
```


All rights reserved

Status CURRENT Effective 8/13/2019


```
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:09:04</DateTimeUtc>
      <Name>Resume From Pause Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:12:34</DateTimeUtc>
      <Name>3602(0,0) High Return Pressure</Name>
      <CategoryCode>2</CategoryCode>
      <OperatorId>FRED</OperatorId>
      <AlarmAlertDetail>
         <AlertAlarmCode>3602</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:12:48</DateTimeUtc>
      <Name>Cleared AlertAlarm.3602</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:12:48</DateTimeUtc>
      <Name>Acknowledged AlertAlarm.</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:12:51</DateTimeUtc>
      <Name>Resume From Pause Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14 :13:02</DateTimeUtc>
      <Name>3602(0,0) High Return Pressure</Name>
      <CategoryCode>2</CategoryCode>
      <OperatorId>FRED</OperatorId>
      <AlarmAlertDetail>
         <AlertAlarmCode>3602</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:05</DateTimeUtc>
   <Name>Cleared AlertAlarm.3602</Name>
   <CategoryCode>8</CategoryCode>
   <OperatorId>FRED</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:05</DateTimeUtc>
   <Name>Acknowledged AlertAlarm.</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>FRED</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:10</DateTimeUtc>
   <Name>Resume From Pause Selected</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>FRED</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:21</DateTimeUtc>
   <Name>3602(0,0) High Return Pressure</Name>
   <CategoryCode>2</CategoryCode>
   <OperatorId>FRED</OperatorId>
   <AlarmAlertDetail>
      <AlertAlarmCode>3602</AlertAlarmCode>
      <AlertAlarmClassCode>4</AlertAlarmClassCode>
      <AlertAlarmClass>Alert</AlertAlarmClass>
      <AlarmData1>0</AlarmData1>
      <AlarmData2>0</AlarmData2>
   </AlarmAlertDetail>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:33</DateTimeUtc>
   <Name>Cleared AlertAlarm.3602</Name>
   <CategoryCode>8</CategoryCode>
   <OperatorId>FRED</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:33</DateTimeUtc>
   <Name>Acknowledged AlertAlarm.</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>FRED</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:38</DateTimeUtc>
   <Name>Changed: Target Return Rate->90</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>FRED</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:13:39</DateTimeUtc>
   <Name>Resume From Pause Selected</Name>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:13:50</DateTimeUtc>
      <Name>3602(0,0) High Return Pressure</Name>
      <CategoryCode>2</CategoryCode>
      <OperatorId>FRED</OperatorId>
      <AlarmAlertDetail>
         <AlertAlarmCode>3602</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:13:51</DateTimeUtc>
      <Name>Cleared AlertAlarm.3602</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:13:51</DateTimeUtc>
      <Name>Acknowledged AlertAlarm.</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:13:55</DateTimeUtc>
      <Name>Changed: Target Return Rate->60</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:13:56</DateTimeUtc>
      <Name>Resume From Pause Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:14:23</DateTimeUtc>
      <Name>Changed: Target Return Rate->50</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>FRED</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:19:23</DateTimeUtc>
      <Name>Changed: Operator ID-></Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <DateTimeUtc>2019-03-22T14:20:28</DateTimeUtc>
      <Name>Auto Draw Rate Decrease</Name>
      <CategoryCode>3</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:20:28</DateTimeUtc>
      <Name>Changed: Target Draw Rate->90</Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:24:28</DateTimeUtc>
      <Name>3602(0,0) High Return Pressure</Name>
      <CategoryCode>2</CategoryCode>
      <AlarmAlertDetail>
         <AlertAlarmCode>3602</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:26:01</DateTimeUtc>
      <Name>Changed: Operator ID->VICTOR</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:26:02</DateTimeUtc>
      <Name>Cleared AlertAlarm.3602</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:26:02</DateTimeUtc>
      <Name>Acknowledged AlertAlarm.</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:26:04</DateTimeUtc>
      <Name>Changed: Target Return Rate->50</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:26:05</DateTimeUtc>
      <Name>Resume From Pause Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <DateTimeUtc>2019-03-22T14:27:42</DateTimeUtc>
      <Name>3508(0,0) No Blood Flow</Name>
      <CategoryCode>2</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
      <AlarmAlertDetail>
         <AlertAlarmCode>3508</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:27:42</DateTimeUtc>
      <Name>Changed: Target Draw Rate->80</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:28:22</DateTimeUtc>
      <Name>Cleared AlertAlarm.3508</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:28:22</DateTimeUtc>
      <Name>Acknowledged AlertAlarm.</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:28:23</DateTimeUtc>
      <Name>Resume From Pause Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:31:00</DateTimeUtc>
      <Name>Auto Draw Rate Decrease</Name>
      <CategoryCode>3</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:31:00</DateTimeUtc>
      <Name>Changed: Target Draw Rate->70</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:32:30</DateTimeUtc>
      <Name>3602(0,0) High Return Pressure</Name>
      <CategoryCode>2</CategoryCode>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <OperatorId>VICTOR</OperatorId>
      <AlarmAlertDetail>
         <AlertAlarmCode>3602</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:33:22</DateTimeUtc>
      <Name>Cleared AlertAlarm.3602</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:33:22</DateTimeUtc>
      <Name>Acknowledged AlertAlarm.</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:33:25</DateTimeUtc>
      <Name>Changed: Target Return Rate->50</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:33:35</DateTimeUtc>
      <Name>Resume From Pause Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:33:44</DateTimeUtc>
      <Name>3602(0,0) High Return Pressure</Name>
      <CategoryCode>2</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
      <AlarmAlertDetail>
         <AlertAlarmCode>3602</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T14:35:22</DateTimeUtc>
      <Name>Cleared AlertAlarm.3602</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>VICTOR</OperatorId>
   </ProcedureEvent>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:35:22</DateTimeUtc>
   <Name>Acknowledged AlertAlarm.</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>VICTOR</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:36:35</DateTimeUtc>
   <Name>Resume From Pause Selected</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>VICTOR</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:36:55</DateTimeUtc>
   <Name>3620(0,0) Low Return Pressure</Name>
   <CategoryCode>2</CategoryCode>
   <OperatorId>VICTOR</OperatorId>
   <AlarmAlertDetail>
      <AlertAlarmCode>3620</AlertAlarmCode>
      <AlertAlarmClassCode>4</AlertAlarmClassCode>
      <AlertAlarmClass>Alert</AlertAlarmClass>
      <AlarmData1>0</AlarmData1>
      <AlarmData2>0</AlarmData2>
   </AlarmAlertDetail>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:36:57</DateTimeUtc>
   <Name>Cleared AlertAlarm.3620</Name>
   <CategoryCode>8</CategoryCode>
   <OperatorId>VICTOR</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:36:57</DateTimeUtc>
   <Name>Acknowledged AlertAlarm.</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>VICTOR</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:36:58</DateTimeUtc>
   <Name>Resume From Pause Selected</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>VICTOR</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T14:41:58</DateTimeUtc>
   <Name>Changed: Operator ID-></Name>
   <CategoryCode>4</CategoryCode>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:28:53</DateTimeUtc>
   <Name>3508(0,0) No Blood Flow</Name>
   <CategoryCode>2</CategoryCode>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <AlarmAlertDetail>
         <AlertAlarmCode>3508</AlertAlarmCode>
         <AlertAlarmClassCode>4</AlertAlarmClassCode>
         <AlertAlarmClass>Alert</AlertAlarmClass>
         <AlarmData1>0</AlarmData1>
         <AlarmData2>0</AlarmData2>
      </AlarmAlertDetail>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:28:53</DateTimeUtc>
      <Name>Changed: Target Draw Rate->60</Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:29:11</DateTimeUtc>
      <Name>Changed: Operator ID->KATIE</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>KATIE</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:29:11</DateTimeUtc>
      <Name>Cleared AlertAlarm.3508</Name>
      <CategoryCode>8</CategoryCode>
      <OperatorId>KATIE</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:29:11</DateTimeUtc>
      <Name>Acknowledged AlertAlarm.</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>KATIE</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:29:13</DateTimeUtc>
      <Name>Resume From Pause Selected</Name>
      <CategoryCode>4</CategoryCode>
      <OperatorId>KATIE</OperatorId>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:34:13</DateTimeUtc>
      <Name>Changed: Operator ID-></Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:37:20</DateTimeUtc>
      <Name>Plasma Volume Reached</Name>
      <CategoryCode>3</CategoryCode>
   </ProcedureEvent>
   <ProcedureEvent>
      <DateTimeUtc>2019-03-22T15:45:37</DateTimeUtc>
      <Name>Changed: Target Return Rate->75</Name>
      <CategoryCode>4</CategoryCode>
   </ProcedureEvent>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:45:52</DateTimeUtc>
         <Name>Changed: Target Return Rate->70</Name>
         <CategoryCode>4</CategoryCode>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:53:26</DateTimeUtc>
         <Name>Changed: Operator ID->SARAH</Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:53:27</DateTimeUtc>
         <Name>Disconnect Donor Task Done</Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:55:05</DateTimeUtc>
         <Name>Remove Plasma Collection Container Task Done</Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
   </ProcedureEvents>
   <Timespans count="2">
      <Timespan>
         <TypeCode>1</TypeCode>
         <StartTime>2019-03-22T13:56:27</StartTime>
         <EndTime>2019-03-22T15:53:27</EndTime>
         <Duration>7020</Duration>
      </Timespan>
      <Timespan>
         <TypeCode>2</TypeCode>
         <StartTime>2019-03-22T13:56:29</StartTime>
         <EndTime>2019-03-22T15:48:47</EndTime>
         <Duration>6738</Duration>
      </Timespan>
   </Timespans>
   <ActiveSWVersion>2.0</ActiveSWVersion>
   <ActiveLanguagePackVersion>0.0</ActiveLanguagePackVersion>
   <Donation>
      <TargetReplacementFluidVolume>500</TargetReplacementFluidVolume>
      <DonationId>DIN00014795</DonationId>
      <OperatorStartId>FRED</OperatorStartId>
      <OperatorEndId>SARAH</OperatorEndId>
      <PhlebotomistStartId>MARK</PhlebotomistStartId>
      <PhlebotomistEndId>SARAH</PhlebotomistEndId>
      <WBVolumeProcessed>3290</WBVolumeProcessed>
      <TotalCycles>13</TotalCycles>
      <ACUsed>212</ACUsed>
      <ACRatio>6:100</ACRatio>
      <SalineUsed>500</SalineUsed>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
            <ProcedureProducts count="1">
               <ProcedureProduct>
                  <ProductSubComponents count="1">
                     <ProductSubComponent>
                        <Code>1</Code>
                        <Type>Anticoagulated Plasma</Type>
                        <IsPrimary>true</IsPrimary>
                        <VolumeData>
                           <TargetVolume>880</TargetVolume>
                           <Volume>880</Volume>
                        </VolumeData>
                     </ProductSubComponent>
                  </ProductSubComponents>
               </ProcedureProduct>
            </ProcedureProducts>
         </Donation>
      </ProcedureData>
      <CompleteStatusCode>1</CompleteStatusCode>
      <CompleteStatusName>Procedure Complete</CompleteStatusName>
   </ProcedureRecord>
</DMSMessage>
```

Electronic Record Example for Non–Donation Record

Following is an example of complete Non–Donation Procedure Record exchange between DXT and the DMS, for Aurora Xi instrument with software version 2.0.

```
POST /procedure-record HTTP/1.1
Connection: keep-alive,Keep-Alive
Date: Wed, 27 Mar 2019 04:01:03 GMT
FK-SourceId: 0
FK-Protocol-Version: 3.0.0.0
Cache-Control: no-transform, no-cache
Content-Type: text/xml; charset=utf-8
Host: 172.16.10.23:8060
Content-Length: 6270

<?xml version="1.0"?>
<DMSMessage>
   <ProcedureRecord device="3">
      <ProcedureData>
         <ProcedureRecordKey>20190322105506_BC09933_PR</ProcedureRecordKey>
         <InstrumentTypeCode>3</InstrumentTypeCode>
         <InstrumentSerialNumber>BC09933</InstrumentSerialNumber>
         <InstrumentAdministrativeSettingsSignature>1446836972A59C2B
         </InstrumentAdministrativeSettingsSignature>
         <StartTimeUtc>2019-03-22T15:55:06</StartTimeUtc>
         <EndTimeUtc>2019-03-22T15:55:24</EndTimeUtc>
         <PlasmaRemovedTimeUtc>0001-01-01T00:00:00</PlasmaRemovedTimeUtc>
         <RecordLastUpdateTimeUtc>2019-03-22T15:55:25</RecordLastUpdateTimeUtc>
         <TimeZone>-05:00</TimeZone>
         <ProcedureDeviceControllers count="3">
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
      <ProcedureDeviceController>
         <Name>Hardware Controller SW Version</Name>
         <Version>10</Version>
      </ProcedureDeviceController>
      <ProcedureDeviceController>
         <Name>MPU SW Version</Name>
         <Version>2.0.18.109</Version>
      </ProcedureDeviceController>
      <ProcedureDeviceController>
         <Name>Safety Controller SW Version</Name>
         <Version>1</Version>
      </ProcedureDeviceController>
   </ProcedureDeviceControllers>
   <RecordType>3</RecordType>
   <ProcedureTypeCode>11</ProcedureTypeCode>
   <ProcedureType>Plasma Collection, Saline</ProcedureType>
   <ProcedureEvents count="16">
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:55:08</DateTimeUtc>
         <Name>Changed: Operator ID->SARAH</Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:55:23</DateTimeUtc>
         <Name>Remove Disposable Set Task Done</Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:55:23</DateTimeUtc>
         <Name>New Procedure Selected</Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
         <Name>Changed: Target Plasma Collection Volume->200</Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
         <Name>Changed: Donor ID-></Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
      </ProcedureEvent>
      <ProcedureEvent>
         <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
         <Name>Changed: Procedure ID-></Name>
         <CategoryCode>4</CategoryCode>
         <OperatorId>SARAH</OperatorId>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019


```
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Donation Setup ID-></Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: ACRatio->6.00000</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Selected Protocol->0.00000</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Target Saline Infusion Volume->500</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Cuff Pressure Draw Setting->52.00000</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Target Draw Rate->120</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Target Return Rate->150</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Minimum Draw Pressure->-250</Name>
   <CategoryCode>4</CategoryCode>
   <OperatorId>SARAH</OperatorId>
</ProcedureEvent>
<ProcedureEvent>
   <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
   <Name>Changed: Maximum Return Pressure->300</Name>
```


Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
            <CategoryCode>4</CategoryCode>
            <OperatorId>SARAH</OperatorId>
         </ProcedureEvent>
         <ProcedureEvent>
            <DateTimeUtc>2019-03-22T15:55:24</DateTimeUtc>
            <Name>Changed: Squeeze Threshold->-110</Name>
            <CategoryCode>4</CategoryCode>
            <OperatorId>SARAH</OperatorId>
         </ProcedureEvent>
      </ProcedureEvents>
      <ActiveSWVersion>2.0</ActiveSWVersion>
      <ActiveLanguagePackVersion>0.0</ActiveLanguagePackVersion>
      <NonDonation>
         <TargetReplacementFluidVolume>500</TargetReplacementFluidVolume>
         <OperatorStartId>SARAH</OperatorStartId>
         <OperatorEndId>SARAH</OperatorEndId>
         <ProcedureSoftGoods count="0" />
      </NonDonation>
   </ProcedureData>
   <CompleteStatusCode>1</CompleteStatusCode>
   <CompleteStatusName>Procedure Complete</CompleteStatusName>
</ProcedureRecord>
</DMSMessage>
```

Section 3.2 Procedure Parameter Status

DXT will use the Procedure Parameter Status service to notify DMS on the status of the procedure parameter, if DXT is licensed for the remote procedure setup.

Procedure Parameter Status URL

```
POST {Procedure Parameters URL}/procedure-parameter-status HTTP/1.1
```

DXT will send procedure parameter statuses to the address configured as “Procedure Parameters URL” through DXT User Interface.

ProcedureParameterStatus XSD Definition

This complex type defines the contents of the Procedure Parameters Status structure as shown below.

```
<xsd:complexType name="ProcedureParameterStatus">
    <xsd:sequence>
        <xsd:element name="DonationSetupId">
            <xsd:simpleType>
                <xsd:restriction base="xsd:string">
                    <xsd:minLength value="1" />
                    <xsd:maxLength value="25" />
                </xsd:restriction>
            </xsd:simpleType>
        </xsd:element>
```


Fresenius Kabi USA, LLC
All rights reserved

```
        <xsd:element name="ProcedureParametersStatusCode" type="xsd:unsignedInt" />

        <xsd:element name="ValidateFailureDetail" type="xsd:string" minOccurs="0" />
    </xsd:sequence>
</xsd:complexType>
```

ProcedureParameterStatus Data Description

The following table explains the content of ProcedureParameterStatus:

Element	Parent Element	Data Type	Description
ProcedureParameterStatus	DMSMessage		Container. This tag is required.
DonationSetupId	ProcedureParameterStatus	Alphanumeric string. 25 characters max.	Same as the DonationSetupId field, which was sent from DMS in ProcedureParameter message. This element is always provided.
ProcedureParametersStatusCode	ProcedureParameterStatus	Unsigned integer	Parameters status: 1 – parameters were received by the instrument; 2 – parameters were accepted by the instrument; 3 – parameters were rejected by the instrument. More information may be provided in ValidateFailureDetail element; 4 – parameters were accepted and/or confirmed by the operator at the instrument; 5 – parameters were rejected by the operator at the instrument; 8 – instrument is not in a valid state to accept parameters or is not configured to accept parameters in the push mode. If neither confirmation questions nor questionnaire were supplied in procedure setup file, the status will be set to 4 after successful validation by the instrument. This element is always provided.

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Element	Parent Element	Data Type	Description
ValidateFailureDetail	ProcedureParameterStatus	string	Description of the reason for instrument rejection of the parameters. This element is provided only with Status Code 3.

ProcedureParameterStatus Example

Following is an example of complete Procedure Parameter Status between DXT and the DMS.

```
POST /procedure-parameter-status HTTP/1.1
Connection: keep-alive,Keep-Alive
Date: Tue, 30 Oct 2018 16:11:22 GMT
FK-SourceId: 0
FK-DestinationId: 1
FK-Protocol-Version: 3.0.0.0
Cache-Control: no-transform, no-cache
Content-Type: text/xml; charset=utf-8
Host: 127.0.0.1:8060
Content-Length: 220

<DMSMessage>
    <ProcedureParameterStatus>
        <DonationSetupId>14EILB093031591</DonationSetupId>
        <ProcedureParametersStatusCode>2</ProcedureParametersStatusCode>
    </ProcedureParameterStatus>
</DMSMessage>
```

Section 3.3 DMS Response

This section lists responses that DXT expects from the DMS.

OK Response

Upon successful processing, the DMS should respond to with the following:

```
Header: HTTP 1.1 200 OK
Content-Length: 0 Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0
```

Error Response

When the DMS encounters problems validating the request header, the response to DXT should be formatted as following:

```
HTTP/1.1 {HTTP_ERROR_CODE} {HTTP_ERROR}
```

Status CURRENT Effective 8/13/2019

Fresenius Kabi USA, LLC
All rights reserved

Status CURRENT Effective 8/13/2019

```
Content-Length: XML_LENGTH
Content-Type: text/xml; charset=utf-8
Date: TIMESTAMP
Cache-Control: no-cache no-transform
FK-SourceId: DATA_SOURCE
FK-DestinationId: 0
FK-Protocol-Version: 3.0.0.0

<DXTMessage>
    <ErrorResponse>
        <Code>DMS_ERROR_CODE</Code>
        <Description>DMS_ERROR_DESCRIPTION</Description>
    </ErrorResponse>
</DXTMessage>
```

DXT will treat any non–OK response as an error response. DXT does not specifically act on the type of the error response.

When DXT receives any error response to any message, it will retry sending the message up to three times using the delay programmed for each DMS on DXT User Interface between each retry.

When sending Procedure Record, after all retry attempts are exhausted and transmission is still unsuccessful, DXT will record DMS_ERROR_ CODE and DMS_ERROR_DESCRIPTION of last unsuccessful attempt and will display them along with the record on the undelivered records screen as a failure code. If XML message body is not included in DMS error response or correctly formatted ErrorResponse element is not present in XML message, DXT will record and display HTTP_ERROR_ CODE and HTTP_ERROR instead. Refer to DXT Data Management System User's Guide Chapter 4 – Monitoring for an additional information on handling undelivered records.

When sending Procedure Parameter Status, after all retry attempts are exhausted and transmission is still unsuccessful, DXT will abandon transmission and no further attempts to resend status message will be made.

Over time, this information may be subject to change based upon modification in product manufacturing, blood banking, or critical practice. For additional questions please contact the Clinical Technical Hotline at 1–800–937–5060.

Fresenius Kabi and DXT are trademarks of Fresenius Kabi AG.

Fresenius Kabi USA, LLC
All rights reserved